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(54) **INTERACTION BETWEEN IBC AND AFFINE**

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See application file for complete search history.

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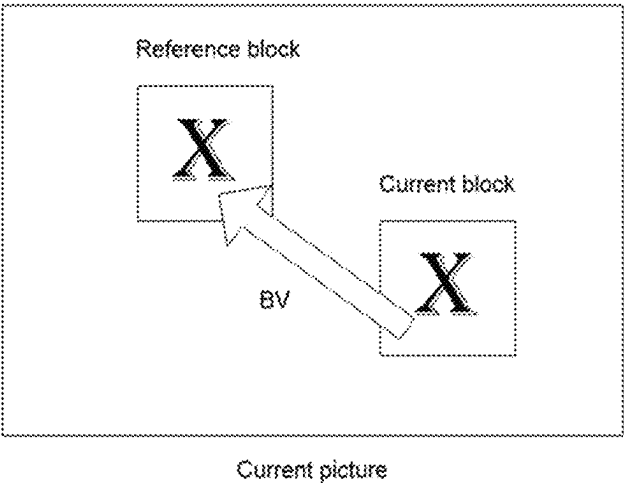
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(57) **ABSTRACT**
Devices, systems and methods for applying intra-block copy (IBC) in video coding are described. In general, methods for integrating IBC with existing motion compensation algorithms for video encoding and decoding are described. In a representative aspect, a method for video encoding using IBC includes determining whether a current block of the current picture is to be encoded using a motion compensation algorithm, and encoding, based on the determining, the current block by selectively applying an intra-block copy to the current block. In a representative aspect, another method for video encoding using IBC includes determining whether a current block of the current picture is to be encoded using an intra-block copy, and encoding, based on the determining, the current block by selectively applying a motion compensation algorithm to the current block.

16 Claims, 21 Drawing Sheets



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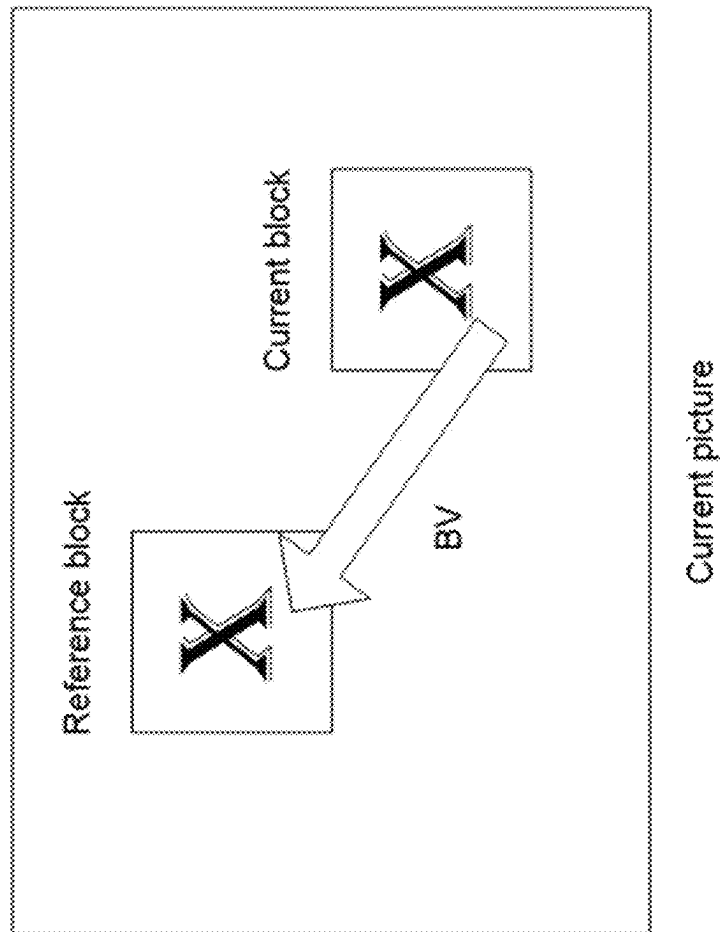


FIG. 1

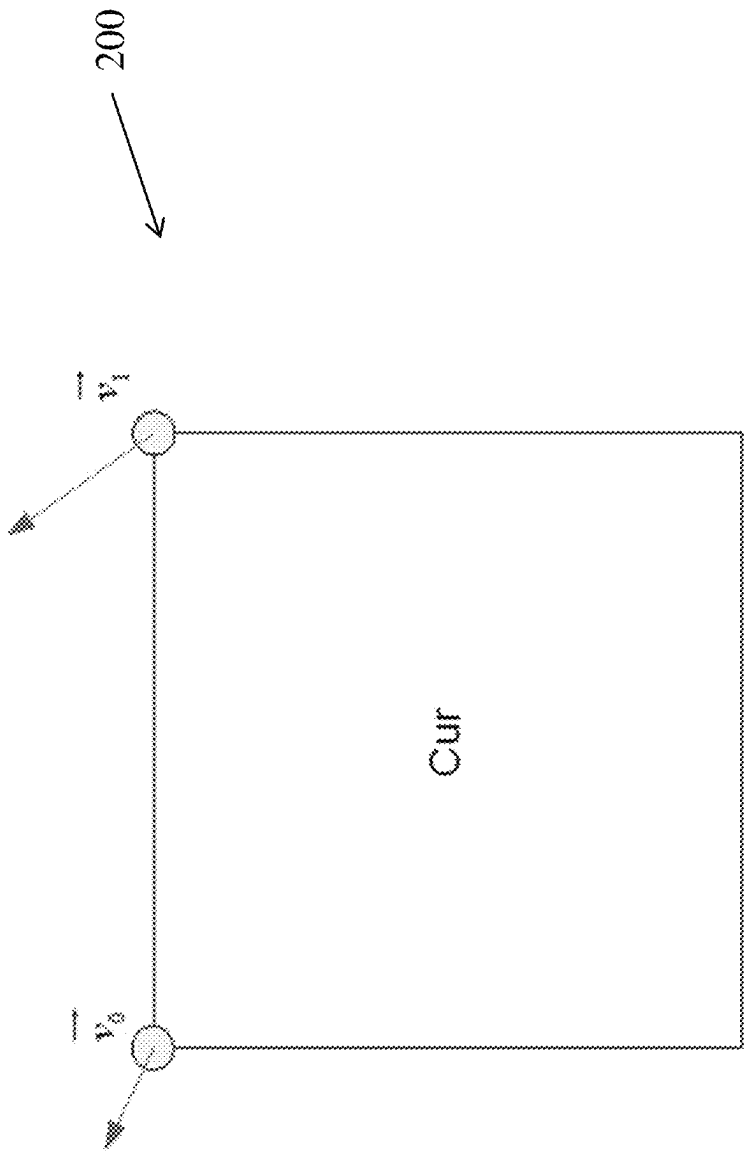


FIG. 2

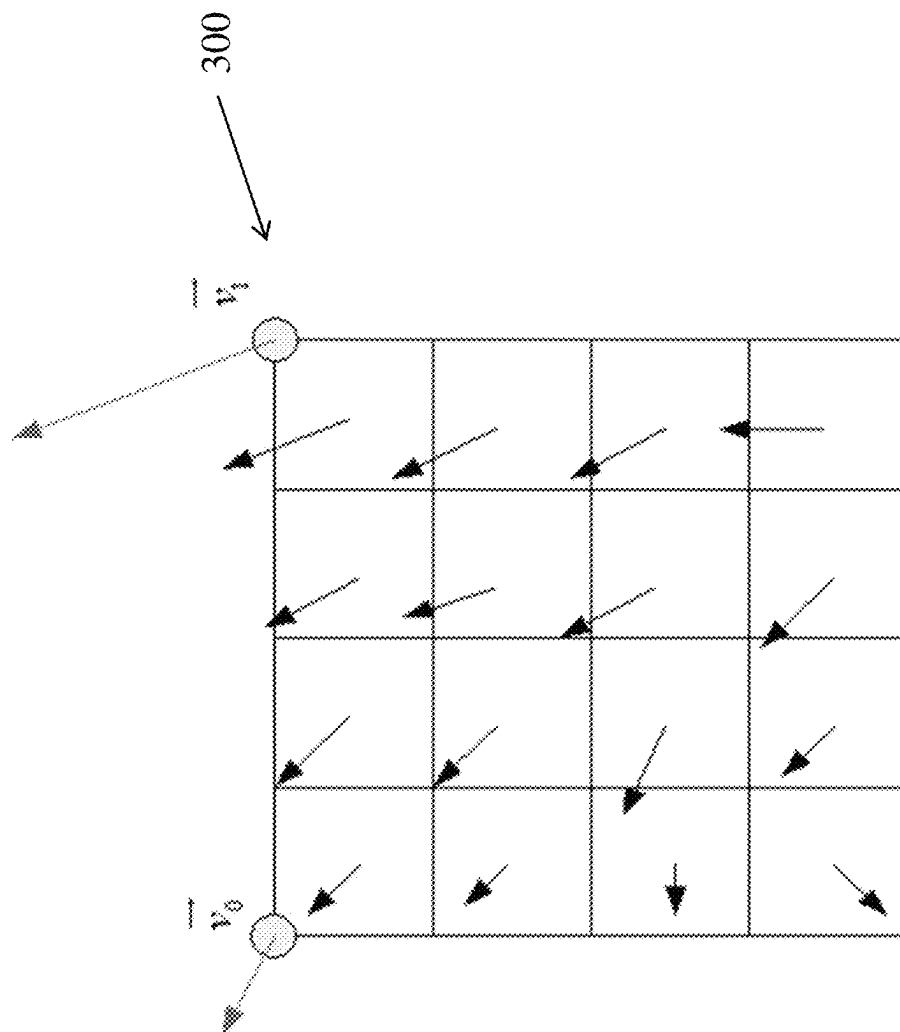


FIG. 3

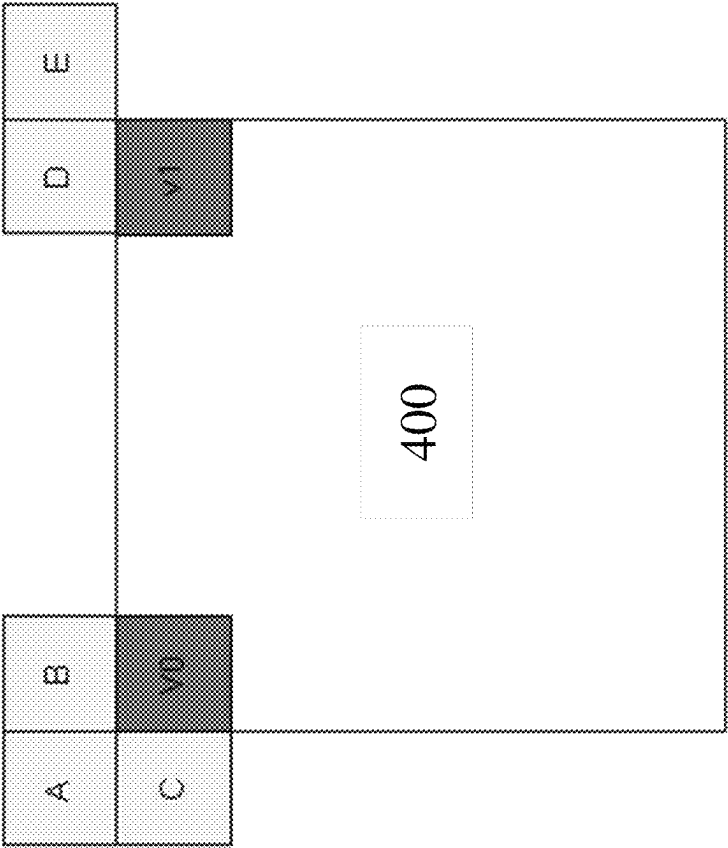


FIG. 4

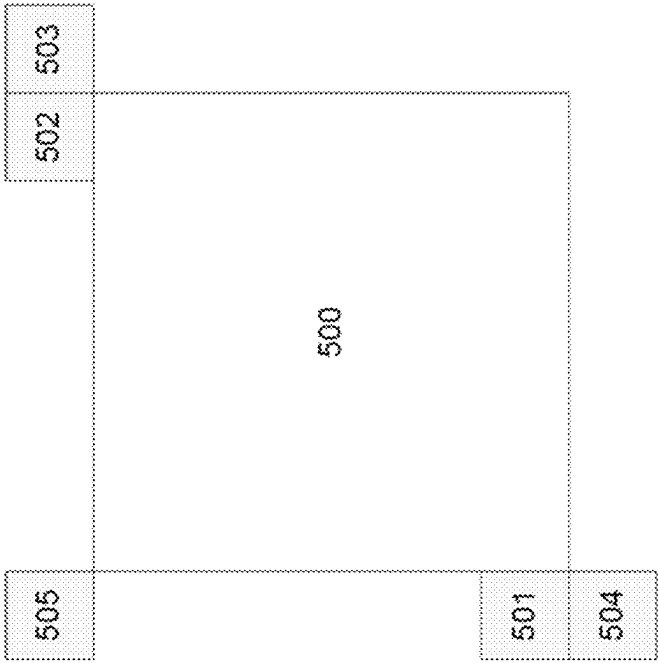


FIG. 5A

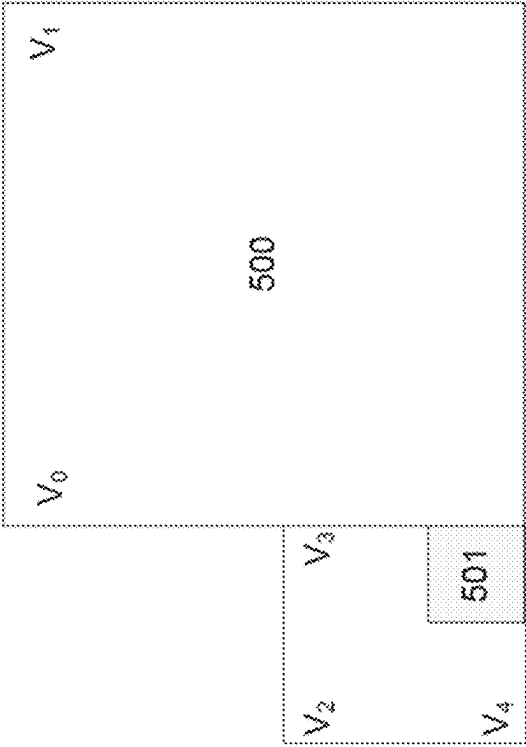


FIG. 5B

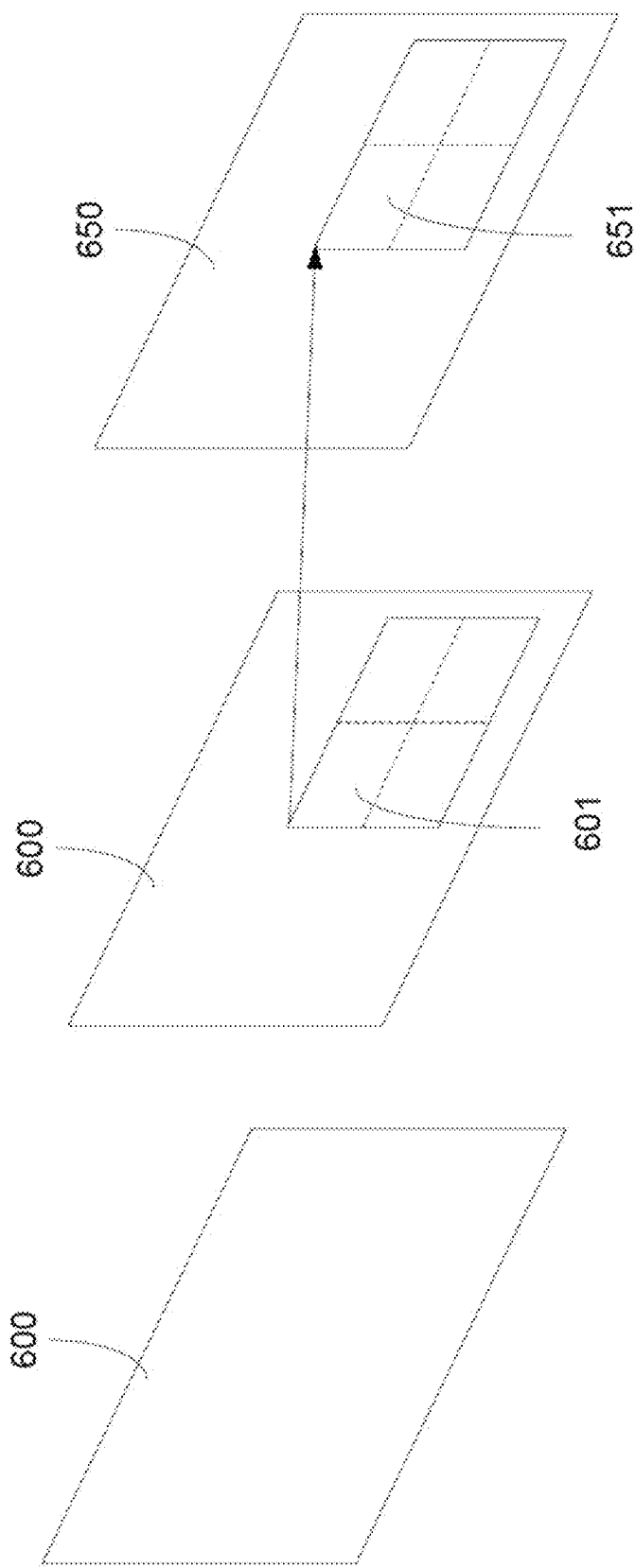


FIG. 6

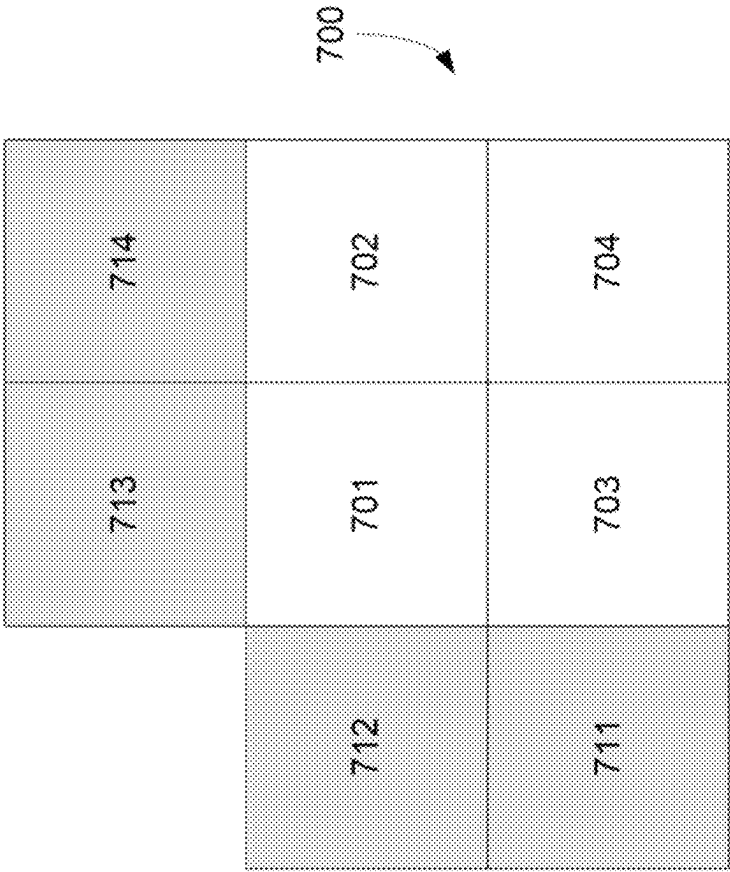


FIG. 7

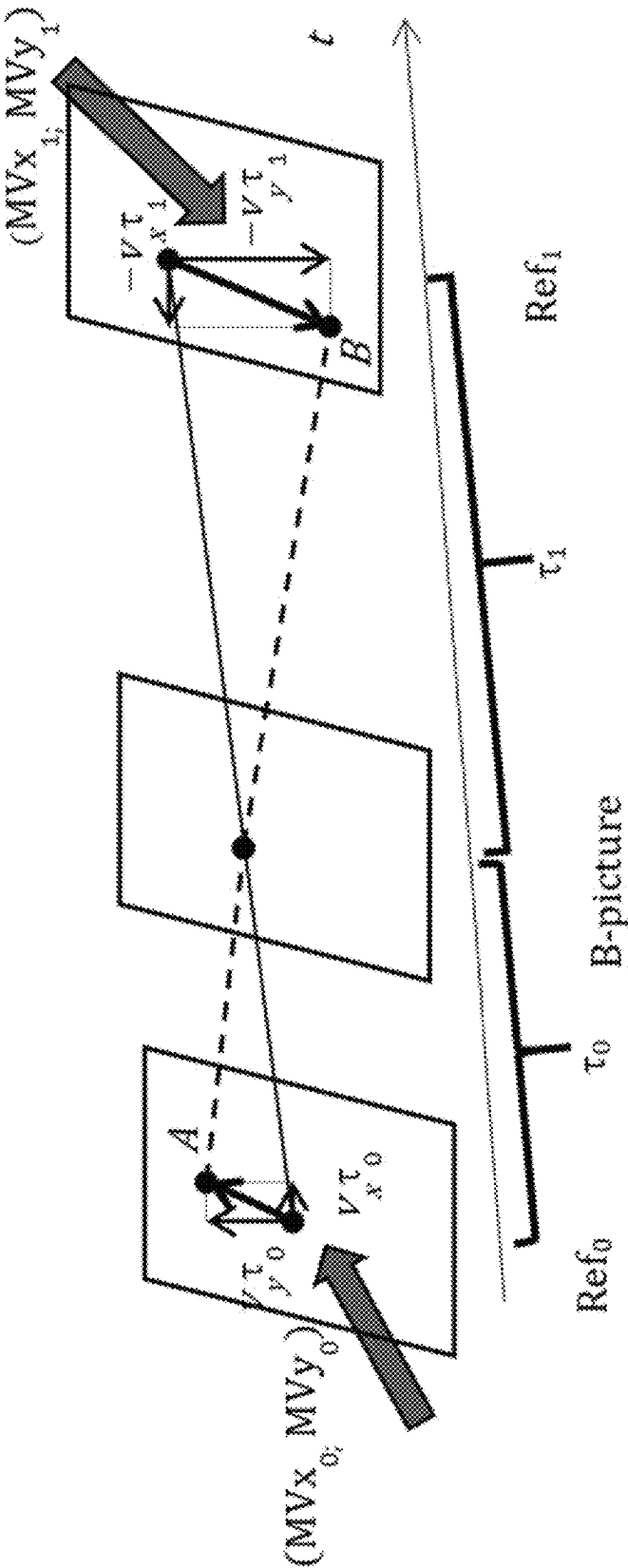


FIG. 8

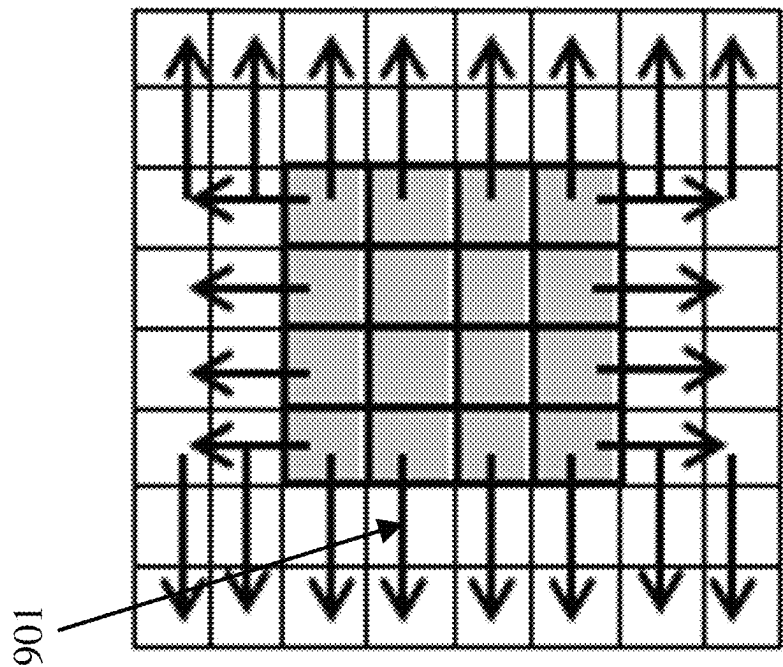


FIG. 9B

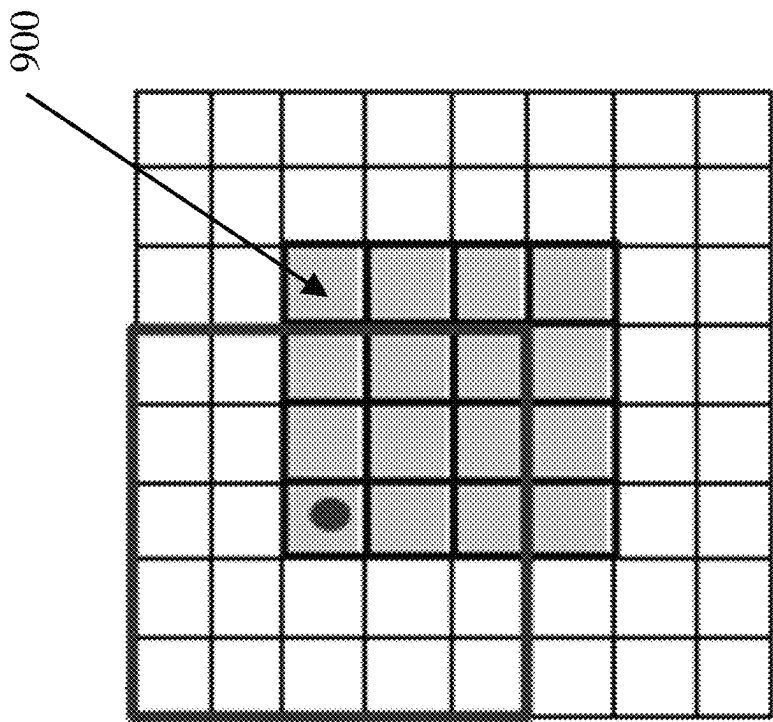


FIG. 9A

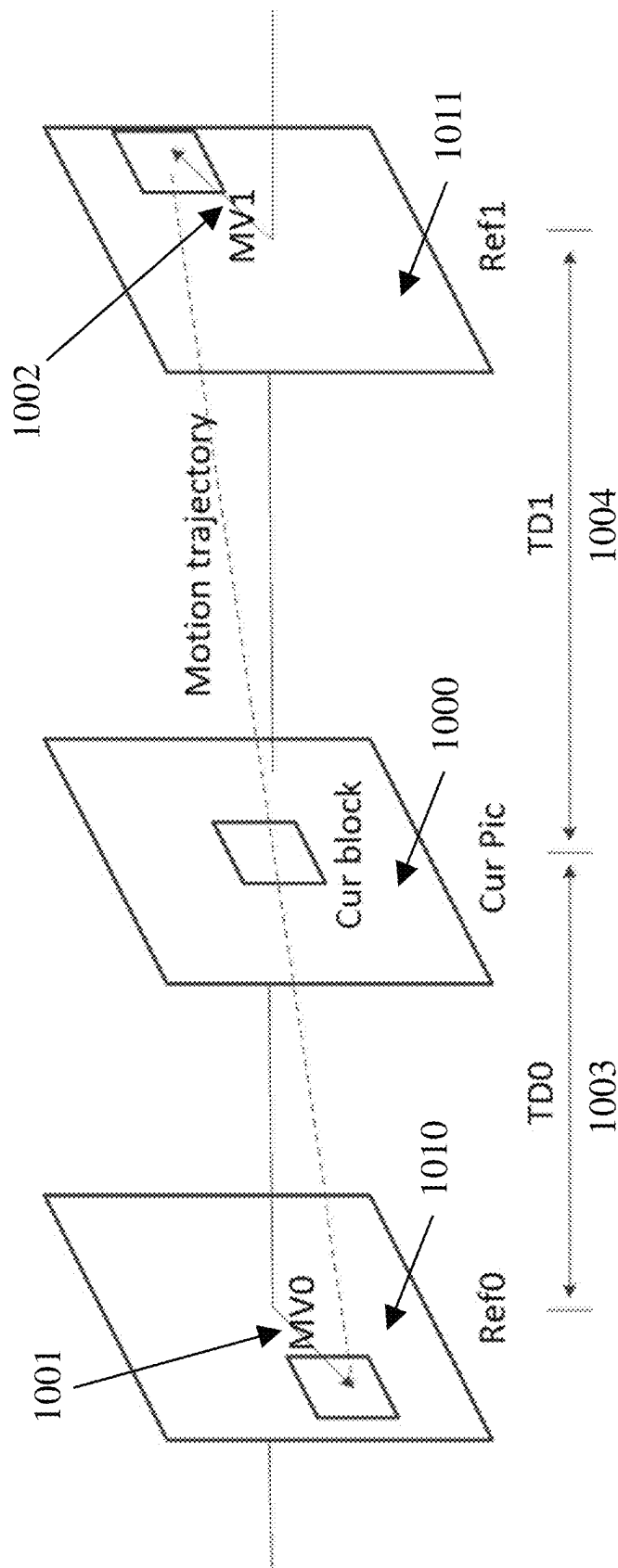


FIG. 10

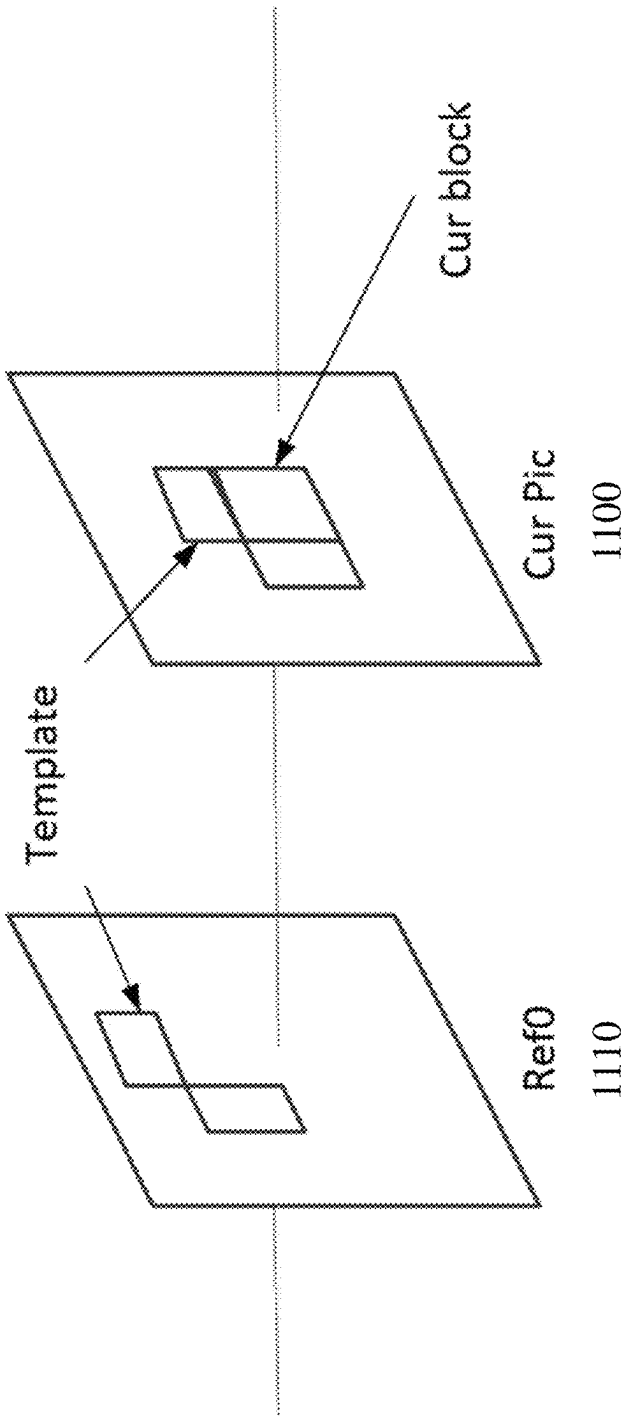


FIG. 11

1200

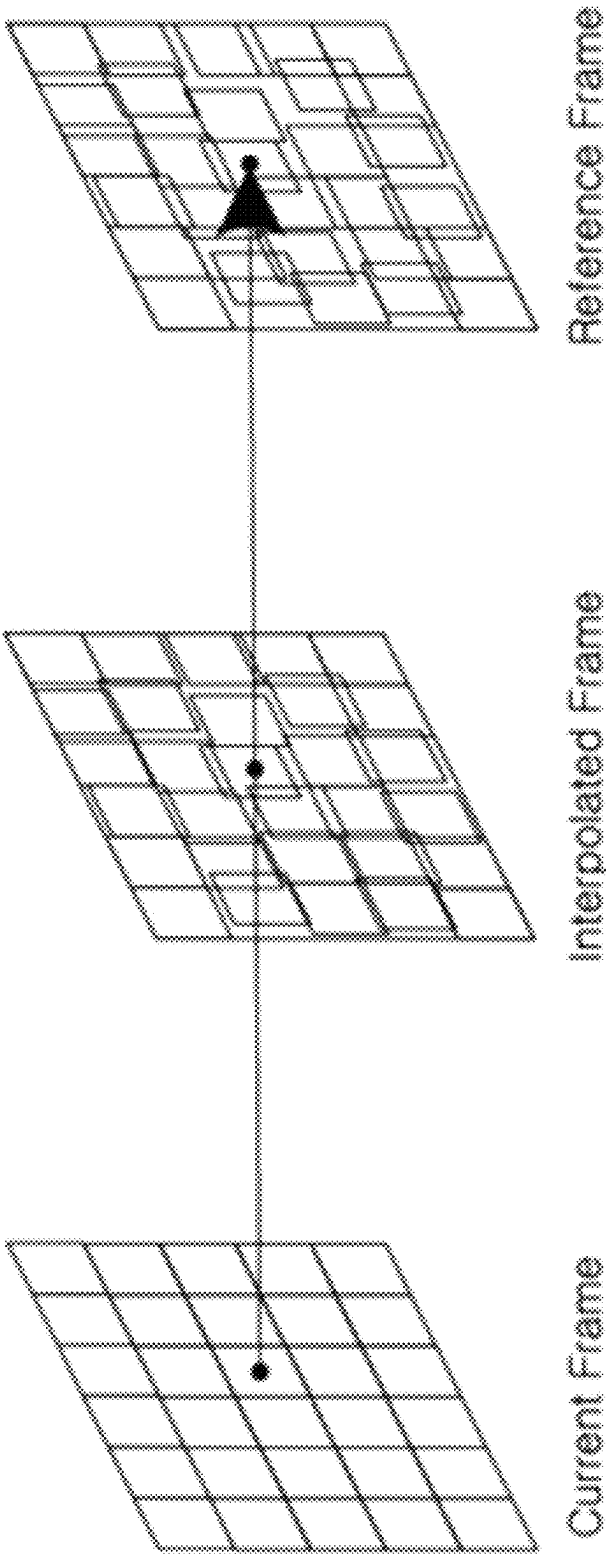


FIG. 12

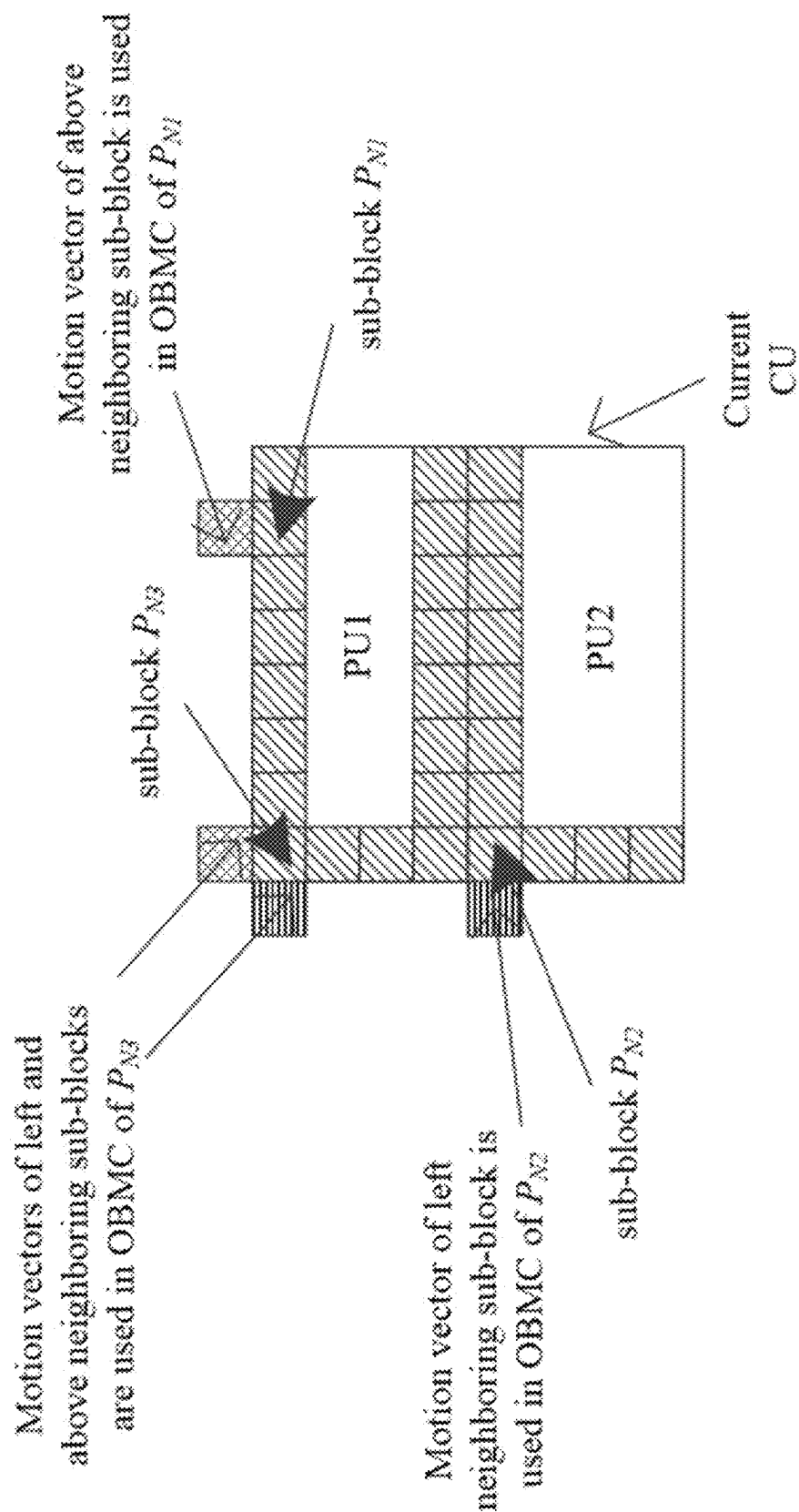


FIG. 13A

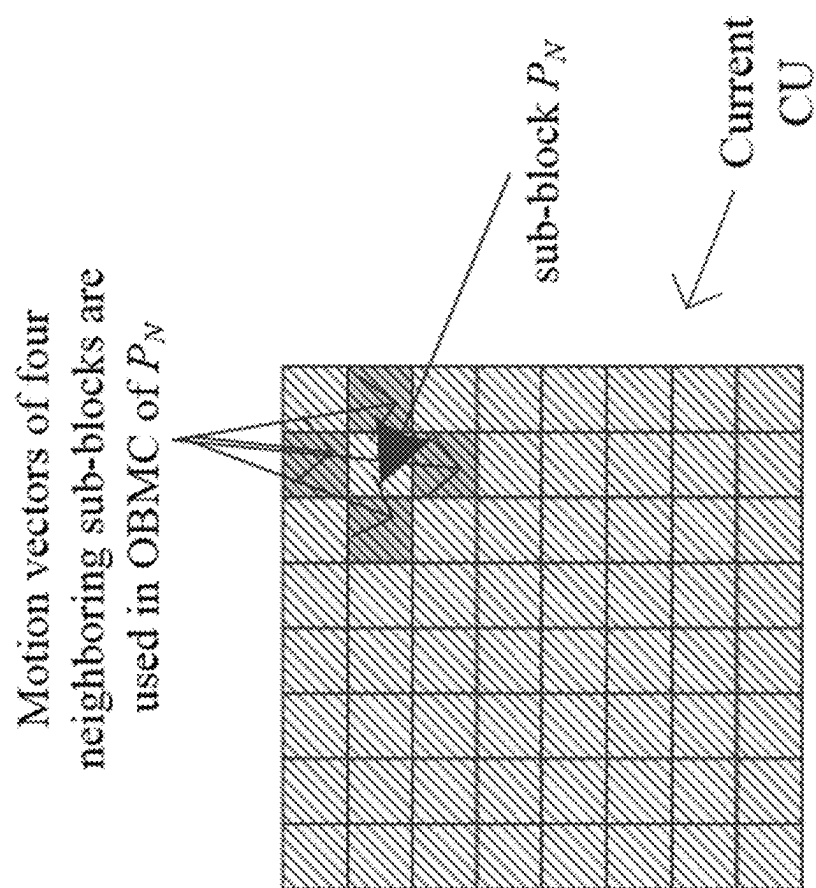


FIG. 13B

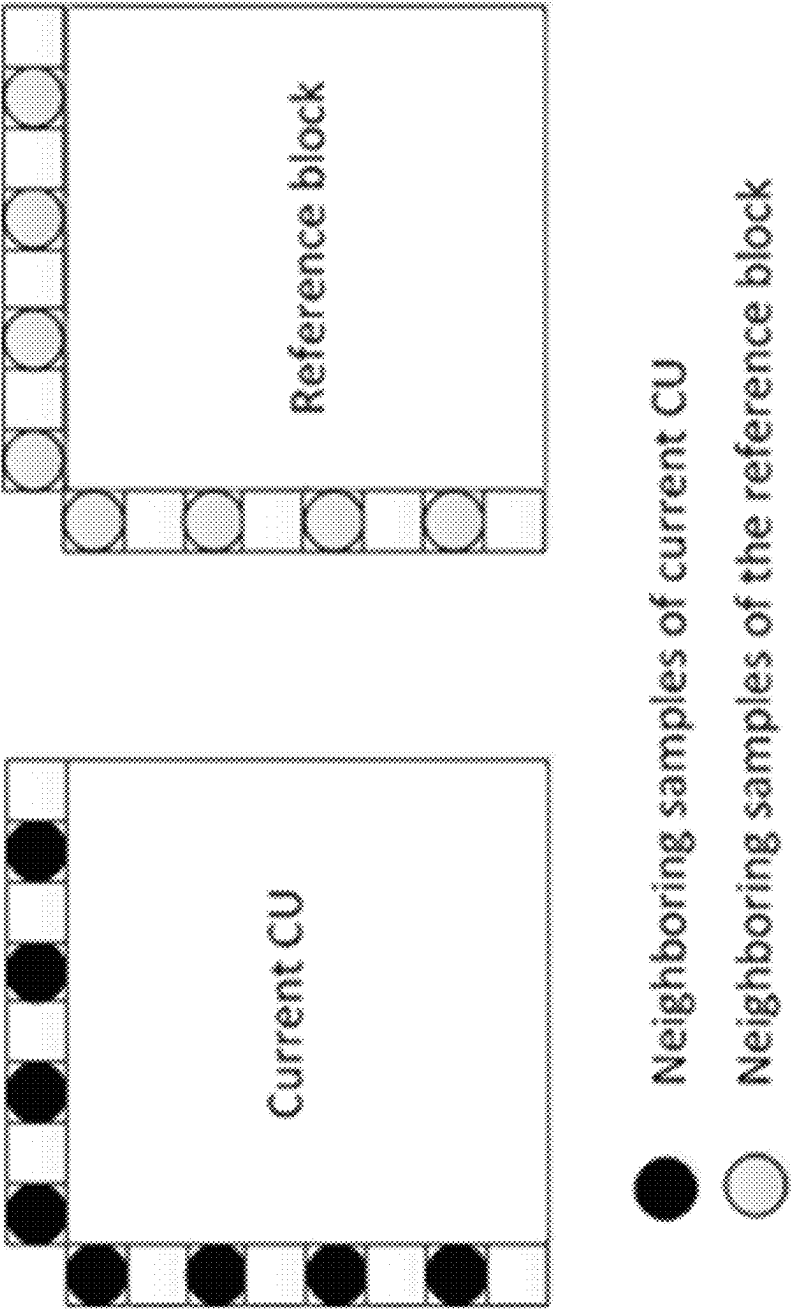


FIG. 14

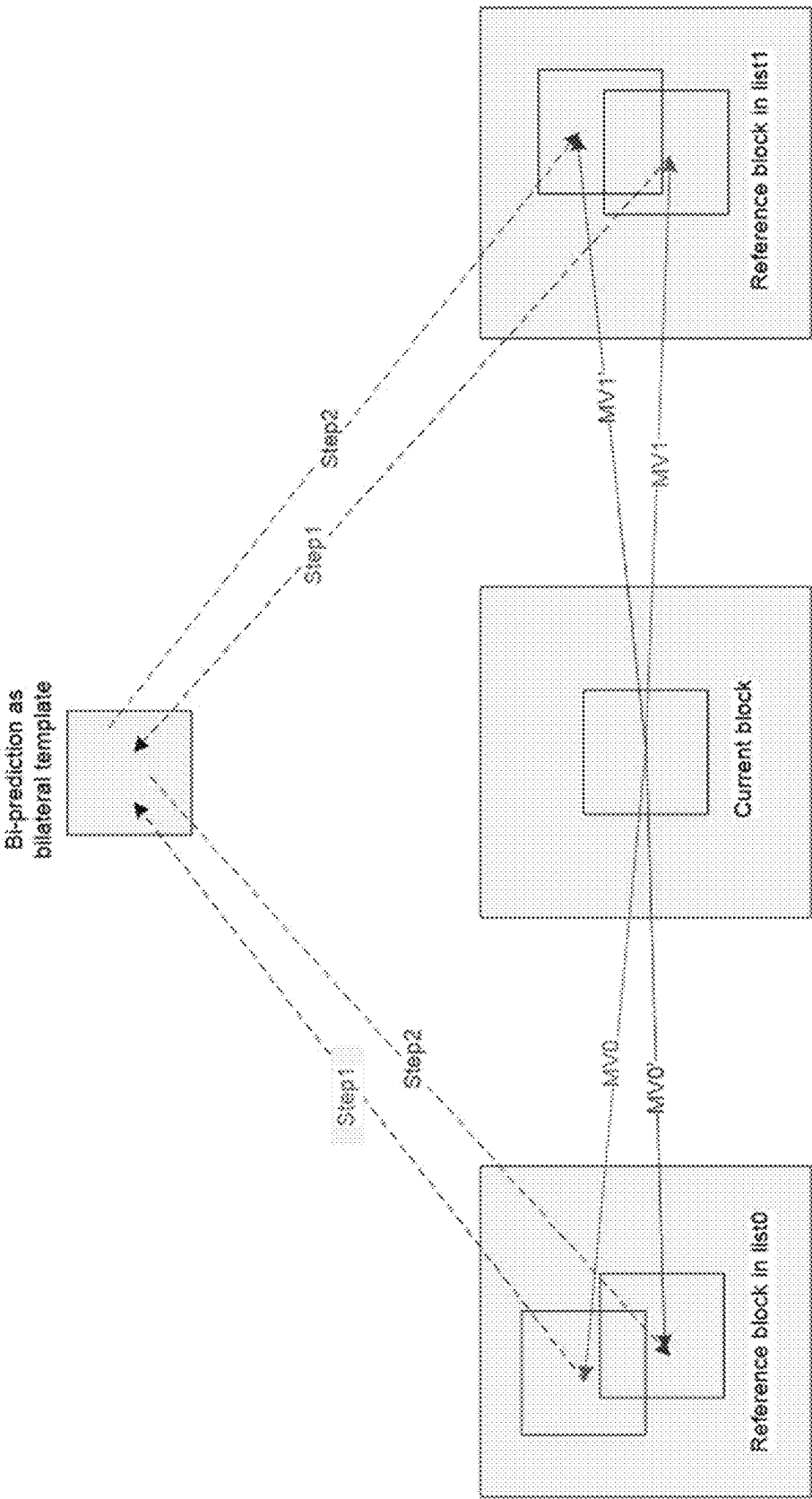


FIG. 15

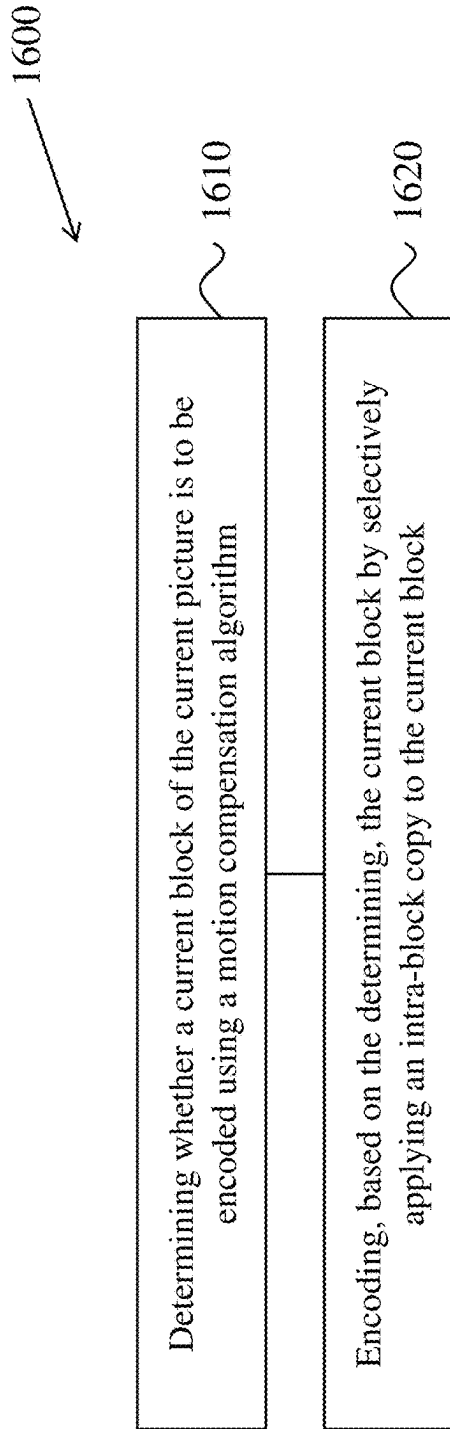


FIG. 16

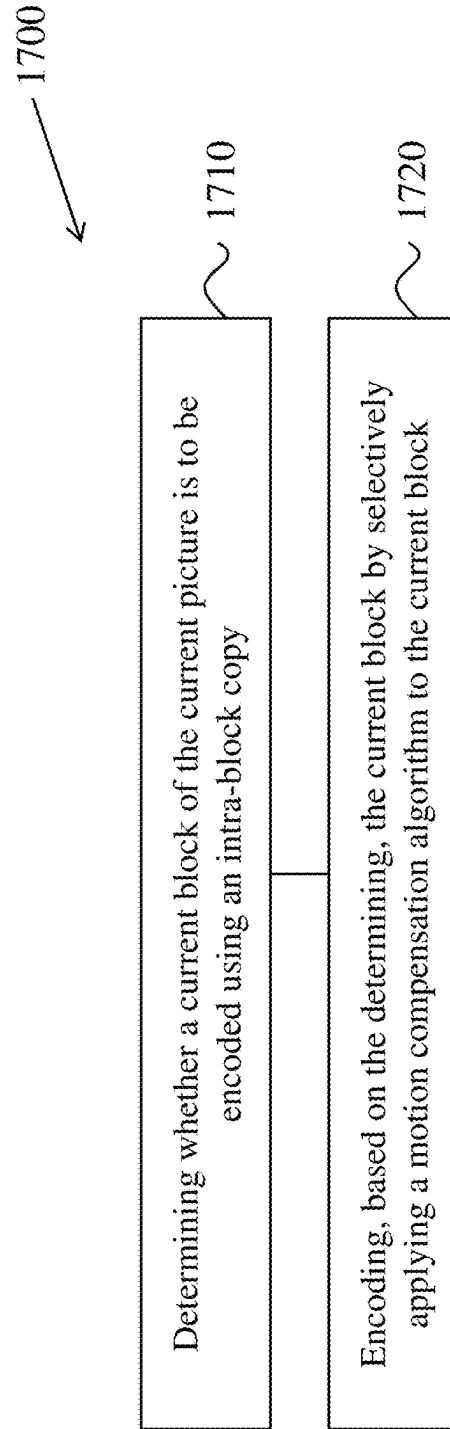


FIG. 17

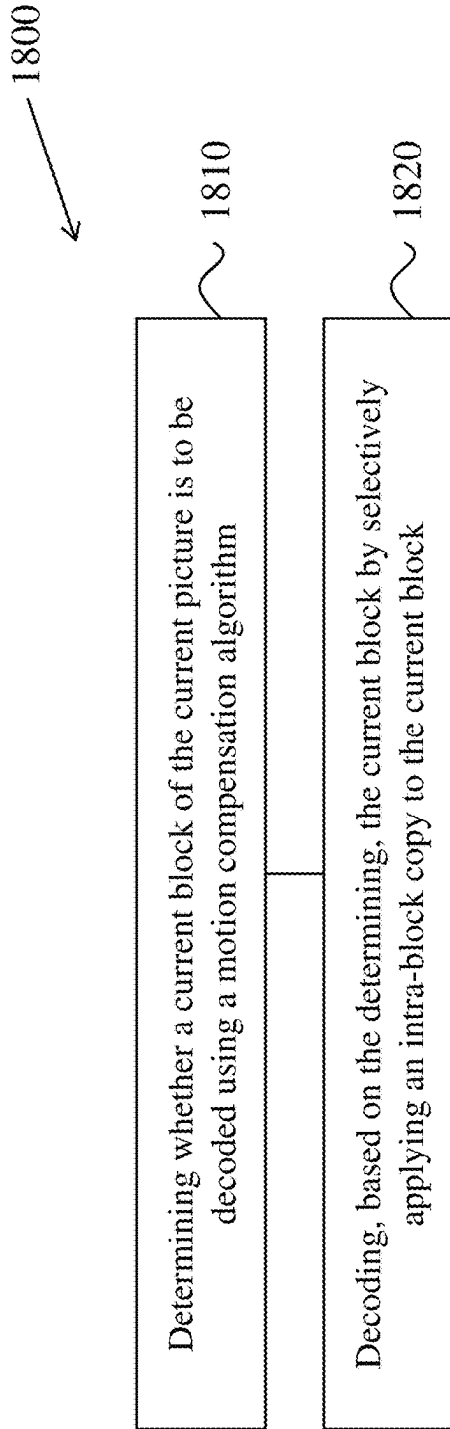


FIG. 18

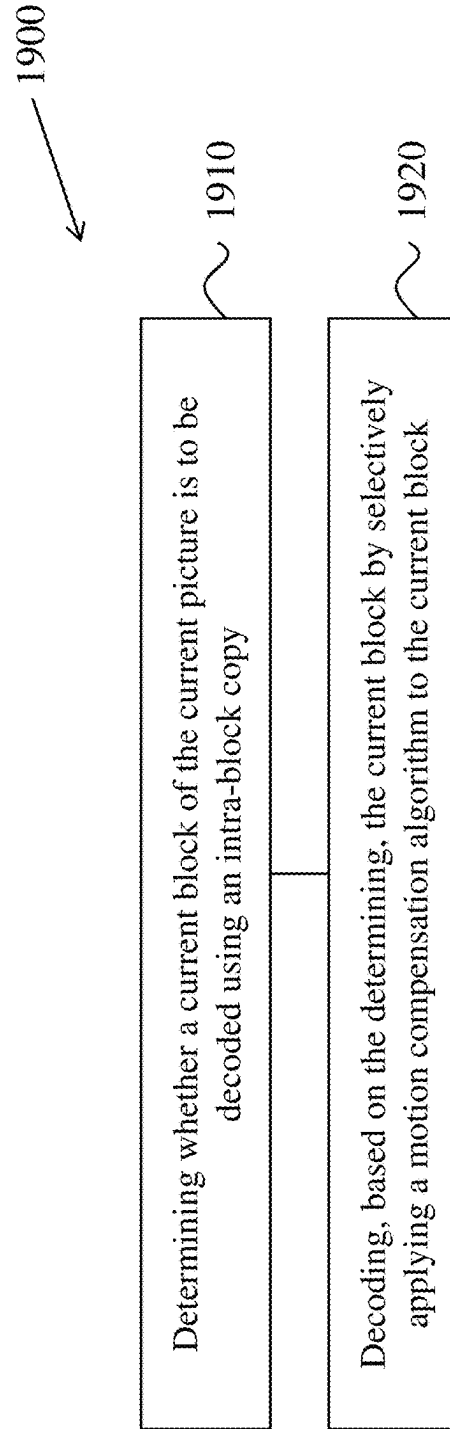


FIG. 19

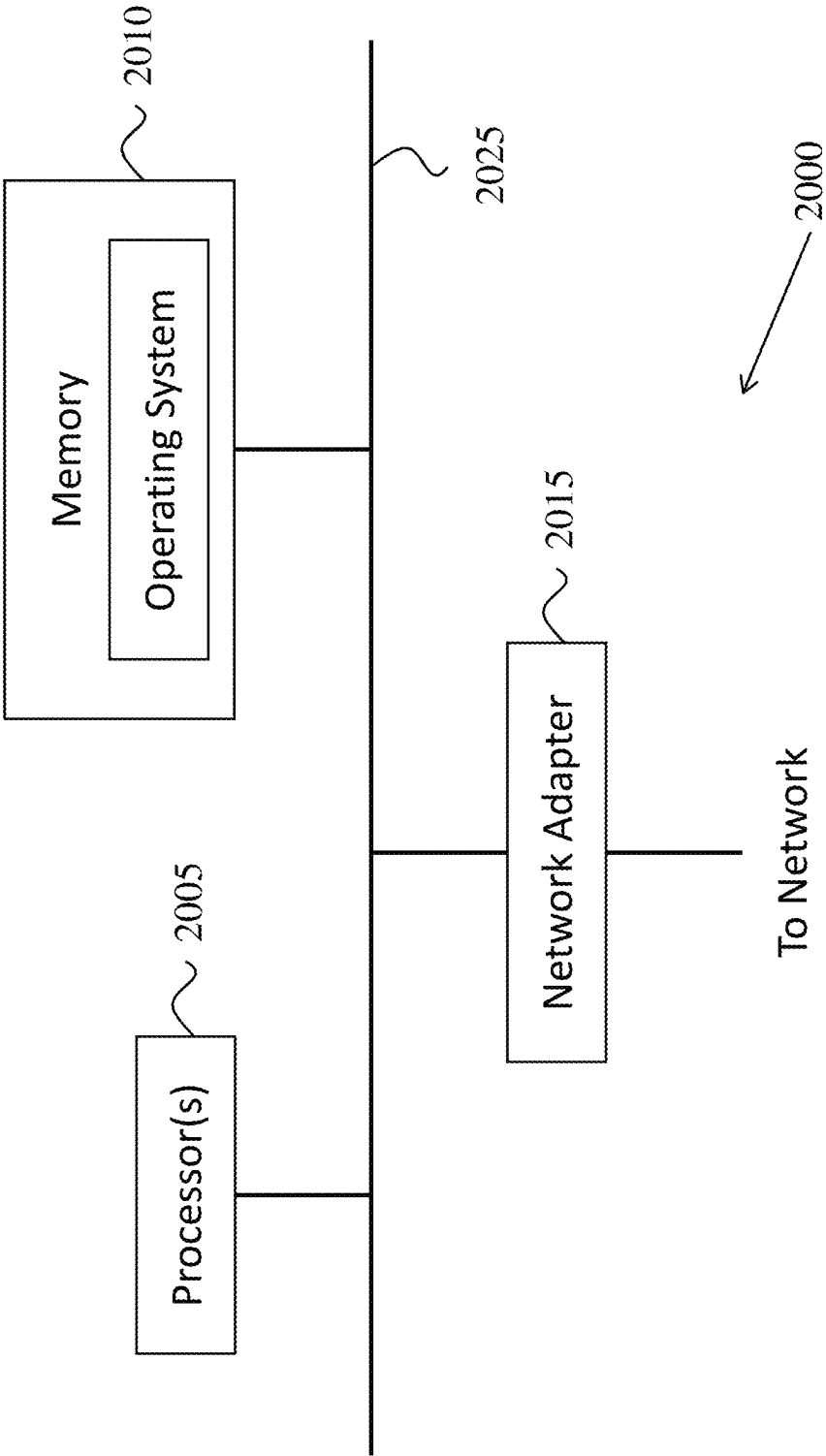


FIG. 20

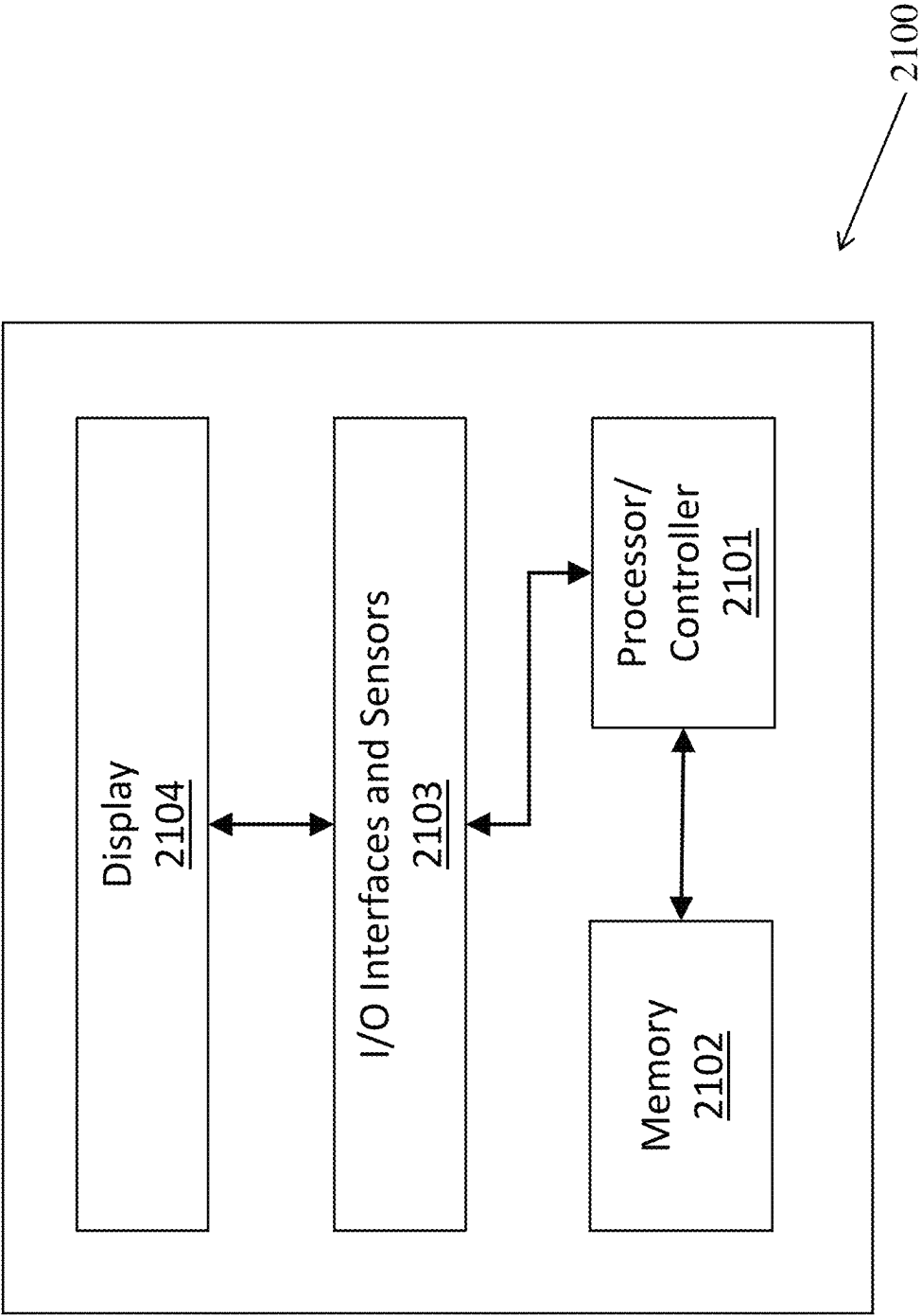


FIG. 21

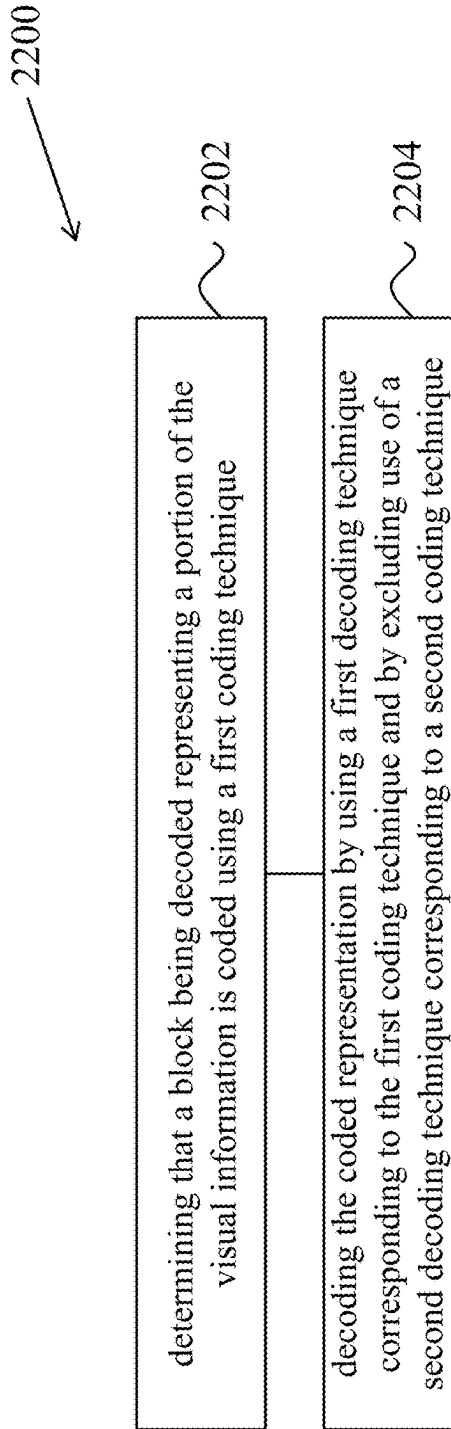


FIG. 22

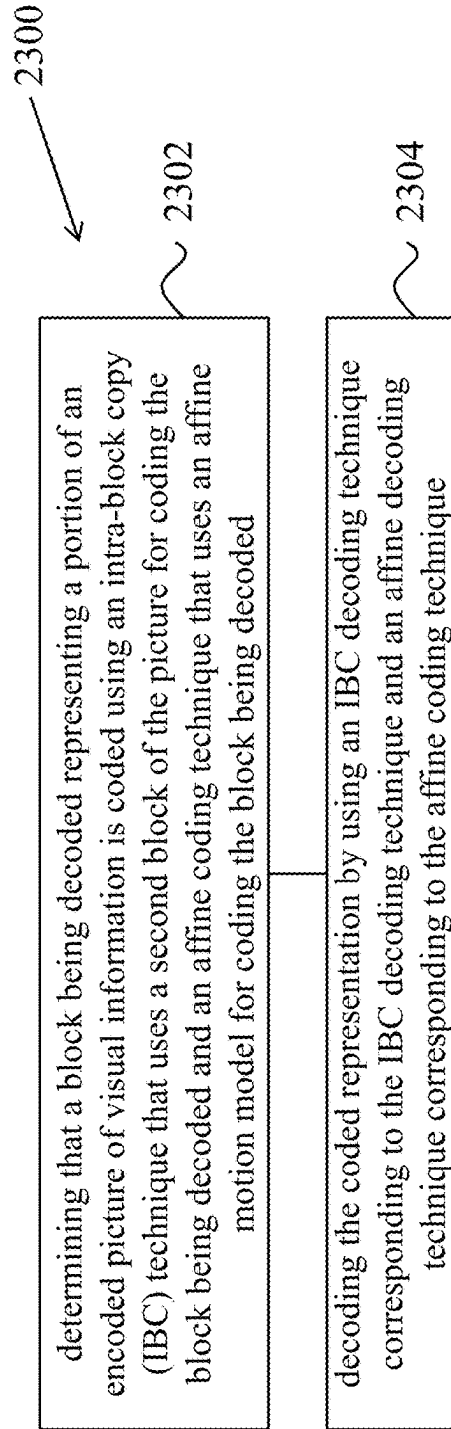


FIG. 23

INTERACTION BETWEEN IBC AND AFFINE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 17/412,771, filed on Aug. 26, 2021, which is a continuation of U.S. application Ser. No. 17/005,521, filed on Aug. 28, 2020, which is a continuation of International Application No. PCT/IB2019/054612, filed on Jun. 4, 2019, which claims the priority to and benefits of International Patent Application No. PCT/CN2018/089920, filed on Jun. 5, 2018. All the aforementioned patent applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

The present disclosure is directed generally to video coding technologies.

BACKGROUND

Motion compensation is a technique in video processing to predict a frame in a video, given the previous and/or future frames by accounting for motion of the camera and/or objects in the video. Motion compensation can be used in the encoding and decoding of video data for video compression.

SUMMARY

Devices, systems and methods related to intra-block copy for motion compensation are described.

In one representative aspect, the disclosed embodiments may be used to provide a method for video encoding using intra-block copy. This method includes determining whether a current block of the current picture is to be encoded using a motion compensation algorithm, and encoding, based on the determining, the current block by selectively applying an intra-block copy to the current block.

In another representative aspect, the disclosed embodiments may be used to provide another method for video encoding using intra-block copy. This method includes determining whether a current block of the current picture is to be encoded using an intra-block copy, and encoding, based on the determining, the current block by selectively applying a motion compensation algorithm to the current block.

In yet another representative aspect, the disclosed embodiments may be used to provide a method for video decoding using intra-block copy. This method includes determining whether a current block of the current picture is to be decoded using a motion compensation algorithm, and decoding, based on the determining, the current block by selectively applying an intra-block copy to the current block.

In yet another representative aspect, the disclosed embodiments may be used to provide another method for video decoding using intra-block copy. This method includes determining whether a current block of the current picture is to be decoded using an intra-block copy, and decoding, based on the determining, the current block by selectively applying a motion compensation algorithm to the current block.

In yet another example aspect, a method of video processing is disclosed. The method includes determining that a block being decoded representing a portion of the visual information is coded using a first coding technique; and decoding the coded representation by using a first decoding

technique corresponding to the first coding technique and by excluding use of a second decoding technique corresponding to a second coding technique; wherein one of the first and second coding techniques corresponds to an intra-block copy (IBC) technique that uses a second block of a same video picture for coding the block being decoded and the other corresponds to an affine coding technique that uses an affine motion model for coding the block being decoded.

In yet another example aspect, a method of visual information processing is disclosed. The method includes determining that a block being decoded representing a portion of an encoded picture of visual information is coded using an intra-block copy (IBC) technique that uses a second block of the picture for coding the block being decoded and an affine coding technique that uses an affine motion model for coding the block being decoded; and decoding the coded representation by using an IBC decoding technique corresponding to the IBC decoding technique and an affine decoding technique corresponding to the affine coding technique.

In yet another representative aspect, the above-described method is embodied in the form of processor-executable code and stored in a computer-readable program medium.

In yet another representative aspect, a device that is configured or operable to perform the above-described method is disclosed. The device may include a processor that is programmed to implement this method.

In yet another representative aspect, a video decoder apparatus may implement a method as described herein.

The above and other aspects and features of the disclosed embodiments are described in greater detail in the drawings, the description and the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 shows an example of an intra-block copy technique.

FIG. 2 shows an example of a simplified affine motion model.

FIG. 3 shows an example of an affine motion vector field (MVF) per sub-block.

FIG. 4 shows an example of motion vector prediction (MVP) for the AF_INTER affine motion mode.

FIGS. 5A and 5B show example candidates for the AF_MERGE affine motion mode.

FIG. 6 shows an example of motion prediction using the alternative temporal motion vector prediction (ATMVP) algorithm for a coding unit (CU).

FIG. 7 shows an example of a coding unit (CU) with sub-blocks and neighboring blocks used by the spatial-temporal motion vector prediction (STMVP) algorithm.

FIG. 8 shows an example of an optical flow trajectory used by the bi-directional optical flow (BIO) algorithm.

FIGS. 9A and 9B show example snapshots of using the bi-directional optical flow (BIO) algorithm without block extensions.

FIG. 10 shows an example of bilateral matching in the frame-rate up conversion (FRUC) algorithm.

FIG. 11 shows an example of template matching in the FRUC algorithm.

FIG. 12 shows an example of unilateral motion estimation in the FRUC algorithm.

FIGS. 13A and 13B show example snapshots of sub-block when using the overlapped block motion compensation (OBMC) algorithm.

FIG. 14 shows an example of neighboring samples used to derive parameters for the local illumination compensation (LIC) algorithm.

FIG. 15 shows an example of the decoder-side motion vector refinement (DMVR) algorithm based on bilateral template matching.

FIG. 16 shows a flowchart of an example method for video encoding using intra-block copy in accordance with the disclosed embodiments.

FIG. 17 shows a flowchart of another example method for video encoding using intra-block copy in accordance with the disclosed embodiments.

FIG. 18 shows a flowchart of an example method for video decoding using intra-block copy in accordance with the disclosed embodiments.

FIG. 19 shows a flowchart of another example method for video decoding using intra-block copy in accordance with the disclosed embodiments.

FIG. 20 is a block diagram illustrating an example of the architecture for a computer system or other control device that can be utilized to implement various portions of the presently disclosed embodiments.

FIG. 21 shows a block diagram of an example embodiment of a mobile device that can be utilized to implement various portions of the presently disclosed embodiments.

FIG. 22 shows a flowchart for an example method of visual information processing.

FIG. 23 shows a flowchart for an example method of visual information processing.

DETAILED DESCRIPTION

Section headings are used in the present disclosure for the ease of understanding and do not limit scope of the technologies and embodiments discussed in each section to just that section.

Due to the increasing demand of higher resolution visual information, such as video, images, three-dimensional scenes, etc., video coding methods and techniques are ubiquitous in modern technology. The embodiments described in this application can apply to various visual information including video, images, three-dimensional scenes, etc. A picture of the visual information can be a frame in a video, a portion of an image, an object in a three-dimensional scene, a portion of the three-dimensional scene, etc. A block can be portion of the picture of the visual information such as a coding unit (CU), a largest coding unit (LCU), a sample, a prediction unit (PU) etc. as described in this application. A sub-block of the visual information can be a PU such as a sub-CU, a sample, etc. The PU can be a pixel, a voxel, or a smallest quantum of resolution of the visual information. Video codecs typically include an electronic circuit or software that compresses or decompresses digital video, and are continually being improved to provide higher coding efficiency. A video codec converts uncompressed video to a compressed format or vice versa. There are complex relationships between the video quality, the amount of data used to represent the video (determined by the bit rate), the complexity of the encoding and decoding algorithms, sensitivity to data losses and errors, ease of editing, random access, and end-to-end delay (latency). The compressed format usually conforms to a standard video compression specification, e.g., the High Efficiency Video Coding (HEVC) standard (also known as H.265 or MPEG-H Part 2), the Versatile Video Coding (VVC) standard to be finalized, or other current and/or future video coding standards.

Embodiments of the present disclosure may be applied to existing video coding standards (e.g., HEVC, H.265) and future standards to improve runtime performance. Section headings are used in the present disclosure to improve

readability of the description and do not in any way limit the discussion or the embodiments (and/or implementations) to the respective sections only.

1. Examples of Reference Pictures and Reference Picture Lists

In HEVC, there are two types of reference pictures: short-term and long-term. A reference picture may be marked as “unused for reference” when it becomes no longer needed for prediction reference. A completely new approach for reference picture management, referred to as reference picture set (RPS) or buffer description has been introduced by HEVC.

The process of marking pictures as “used for short-term reference”, “used for long-term reference”, or “unused for reference” is done using the RPS concept. An RPS is a set of picture indicators that is signaled in each slice header and consists of one set of short-term pictures and one set of long-term pictures. After the first slice header of a picture has been decoded, the pictures in the decoded picture buffer (DPB) are marked as specified by the RPS. The pictures in the DPB that are indicated in the short-term picture part of the RPS are kept as short-term pictures. The short-term or long-term pictures in the DPB that are indicated in the long-term picture part in the RPS are converted to or kept as long-term pictures. And finally, pictures in the DPB for which there is no indicator in the RPS are marked as unused for reference. Thus, all pictures that have been decoded that may be used as references for prediction of any subsequent pictures in decoding order must be included in the RPS.

An RPS consists of a set of picture order count (POC) values that are used for identifying the pictures in the DPB. Besides signaling POC information, the RPS also signals one flag for each picture. Each flag indicates whether the corresponding picture is available or unavailable for reference for the current picture. Note that even though a reference picture is signaled as unavailable for the current picture, it is still kept in the DPB and may be made available for reference later on and used for decoding future pictures.

From the POC information and the availability flag, five lists of reference pictures as shown in Table 1 can be created. The list RefPicSetStCurrBefore consists of short-term pictures that are available for reference for the current picture and have POC values that are lower than the POC value of the current picture. RefPicSetStCurrAfter consist of available short-term pictures with a POC value that is higher than the POC value of the current picture. RefPicSetStFoll is a list that contains all short-term pictures that are made unavailable for the current picture but may be used as reference pictures for decoding subsequent pictures in decoding order. Finally, the lists RefPicSetLtCurr and RefPicSetLtFoll contain long-term pictures that are available and unavailable for reference for the current picture, respectively.

TABLE 1

List of Reference Picture lists			
List name	Long-term or short-term	Availability flag	POC
RefPicSetStCurrBefore	Short-term	Available	Lower
RefPicSetStCurrAfter	Short-term	Available	Higher
RefPicSetStFoll	Short-term	Unavailable	—
RefPicSetLtCurr	Long-term	Available	—
RefPicSetLtFoll	Long-term	Unavailable	—

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1.1 Examples of Short-Term and Long-Term
Reference Pictures

The syntax for the general sequence parameter set is shown below:

	Descriptor
seq_parameter_set_rbsp() {	
sps_video_parameter_set_id	u(4)
sps_max_sub_layers_minus1	u(3)
sps_temporal_id_nesting_flag	u(1)
profile_tier_level(1,	
sps_max_sub_layers_minus1)	
sps_seq_parameter_set_id	ue(v)
chroma_format_idc	ue(v)
if(chroma_format_idc == 3)	
...	
}	
...	
amp_enabled_flag	u(1)
sample_adaptive_offset_enabled_flag	u(1)
pcm_enabled_flag	u(1)
if(pcm_enabled_flag) {	
...	
}	
num_short_term_ref_pic_sets	ue(v)
for(i = 0; i < num_short_term_ref_pic_sets; i++)	
st_ref_pic_set(i)	
long_term_ref_pics_present_flag	u(1)
if(long_term_ref_pics_present_flag) {	
num_long_term_ref_pics_sps	ue(v)
for(i = 0; i < num_long_term_ref_pics_sps;	
i++) {	
lt_ref_pic_poc_lsb_sps[i]	u(v)
used_by_curr_pic_lt_sps_flag[i]	u(1)
}	
sps_temporal_mvp_enabled_flag	u(1)
...	
}	

The syntax for the general slice segment header is shown below:

	Descriptor
slice_segment_header() {	
first_slice_segment_in_pic_flag	u(1)
if(nal_unit_type >= BLA_W_LP && nal_unit_type <=	
RSV_IRAP_VCL23)	
no_output_of_prior_pics_flag	u(1)
slice_pic_parameter_set_id	ue(v)
if(!first_slice_segment_in_pic_flag) {	
if(dependent_slice_segments_enabled_flag)	
dependent_slice_segment_flag	u(1)
slice_segment_address	u(v)
}	
if(!dependent_slice_segment_flag) {	
for(i = 0; i < num_extra_slice_header_bits; i++)	
slice_reserved_flag[i]	u(1)
slice_type	ue(v)
if(output_flag_present_flag)	
pic_output_flag	u(1)
if(separate_colour_plane_flag == 1)	
colour_plane_id	u(2)
if(nal_unit_type != IDR_W_RADL && nal_unit_type	
!= IDR_N_LP) {	
slice_pic_order_cnt_lsb	u(v)
short_term_ref_pic_set_sps_flag	u(1)
if(!short_term_ref_pic_set_sps_flag)	
st_ref_pic_set(num_short_term_ref_pic_sets)	
else if(num_short_term_ref_pic_sets > 1)	
short_term_ref_pic_set_idx	u(v)
if(long_term_ref_pics_present_flag) {	
if(num_long_term_ref_pics_sps > 0)	
num_long_term_sps	ue(v)

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-continued

	Descriptor
num_long_term_pics	ue(v)
for(i = 0; i < num_long_term_sps +	
num_long_term_pics; i++) {	
if(i < num_long_term_sps) {	
if(num_long_term_ref_pics_sps > 1)	
lt_idx_sps[i]	u(v)
} else {	
poc_lsb_lt[i]	u(v)
used_by_curr_pic_lt_flag[i]	u(1)
}	
delta_poc_msb_present_flag[i]	u(1)
if(delta_poc_msb_present_flag[i])	
delta_poc_msb_cycle_lt[i]	ue(v)
}	
...	

The semantics used in the syntax tables above are defined as:

num_short_term_ref_pic_sets specifies the number of st_ref_pic_set()syntax structures included in the SPS. The value of num_short_term_ref_pic_sets shall be in the range of 0 to 64, inclusive.

In some embodiments, a decoder may allocate memory for a total number of num_short_term_ref_pic_sets+1 st_ref_pic_set()syntax structures since there may be a st_ref_pic_set()syntax structure directly signaled in the slice headers of a current picture. A st_ref_pic_set()syntax structure directly signaled in the slice headers of a current picture has an index equal to num_short_term_ref_pic_sets.

long_term_ref_pics_present_flag equal to 0 specifies that no long-term reference picture is used for inter prediction of any coded picture in the coded video sequence (CVS). long_term_ref_pics_present_flag equal to 1 specifies that long-term reference pictures may be used for inter prediction of one or more coded pictures in the CVS.

num_long_term_ref_pics_sps specifies the number of candidate long-term reference pictures that are specified in the SPS. The value of num_long_term_ref_pics_sps shall be in the range of 0 to 32, inclusive.

lt_ref_pic_poc_lsb_sps[i] specifies the picture order count modulo MaxPicOrderCntLsb of the i-th candidate long-term reference picture specified in the SPS. The number of bits used to represent lt_ref_pic_poc_lsb_sps[i] is equal to log2_max_pic_order_cnt_lsb_minus4+4.

used_by_curr_pic_lt_sps_flag[i] equal to 0 specifies that the i-th candidate long-term reference picture specified in the SPS is not used for reference by a picture that includes in its long-term reference picture set (RPS) the i-th candidate long-term reference picture specified in the SPS.

short_term_ref_pic_set_sps_flag equal to 1 specifies that the short-term RPS of the current picture is derived based on one of the st_ref_pic_set()syntax structures in the active SPS that is identified by the syntax element short_term_ref_pic_set_idx in the slice header. short_term_ref_pic_set_sps_flag equal to 0 specifies that the short-term RPS of the current picture is derived based on the st_ref_pic_set()syntax structure that is directly included in the slice headers of the current picture. When num_short_term_ref_pic_sets is equal to 0, the value of short_term_ref_pic_set_sps_flag shall be equal to 0.

short_term_ref_pic_set_idx specifies the index, into the list of the st_ref_pic_set() syntax structures included in the active SPS, of the st_ref_pic_set()syntax structure that is used for derivation of the short-term RPS of the current picture. The syntax element short_term_ref_pic_set_idx is

represented by $\text{Ceil}(\text{Log2}(\text{num_short_term_ref_pic_sets}))$ bits. When not present, the value of `short_term_ref_pic_set_idx` is inferred to be equal to 0. The value of `short_term_ref_pic_set_idx` shall be in the range of 0 to `num_short_term_ref_pic_sets-1`, inclusive.

In some embodiments, the variable `CurrRpsIdx` is derived as follows:

If `short_term_ref_pic_set_sps_flag` is equal to 1, `CurrRpsIdx` is set equal to `short_term_ref_pic_set_idx`.

Otherwise, `CurrRpsIdx` is set equal to `num_short_term_ref_pic_sets`.

`num_long_term_sps` specifies the number of entries in the long-term RPS of the current picture that are derived based on the candidate long-term reference pictures specified in the active SPS. The value of `num_long_term_sps` shall be in the range of 0 to `num_long_term_ref_pics_sps`, inclusive. When not present, the value of `num_long_term_sps` is inferred to be equal to 0.

`num_long_term_pics` specifies the number of entries in the long-term RPS of the current picture that are directly signaled in the slice header. When not present, the value of `num_long_term_pics` is inferred to be equal to 0.

In some embodiments, when `nuh_layer_id` is equal to 0, the value of `num_long_term_pics` shall be less than or equal to `sps_max_dec_pic_buffering_minus_1[Temporand]-NumNegativePics[CurrRpsIdx]-NumPositivePics[CurrRpsIdx]-num_long_term_sps-TwoVersionsOfCurrDecPicFlag`.

`lt_idx_sps[i]` specifies an index, into the list of candidate long-term reference pictures specified in the active SPS, of the *i*-th entry in the long-term RPS of the current picture. The number of bits used to represent `lt_idx_sps[i]` is equal to $\text{Ceil}(\text{Log2}(\text{num_long_term_ref_pics_sps}))$. When not present, the value of `lt_idx_sps[i]` is inferred to be equal to 0. The value of `lt_idx_sps[i]` shall be in the range of 0 to `num_long_term_ref_pics_sps-1`, inclusive.

`poc_lsb_lt[i]` specifies the value of the picture order count modulo `MaxPicOrderCntLsb` of the *i*-th entry in the long-term RPS of the current picture. The length of the `poc_lsb_lt[i]` syntax element is `log2_max_pic_order_cnt_lsb_minus4+4` bits.

`used_by_curr_pic_lt_flag[i]` equal to 0 specifies that the *i*-th entry in the long-term RPS of the current picture is not used for reference by the current picture.

In some embodiments, the variables `PocLsbLq[i]` and `UsedByCurrPicLq[i]` are derived as follows:

If *i* is less than `num_long_term_sps`, `PocLsbLq[i]` is set equal to `lt_ref_pic_poc_lsb_spq[lt_idx_sps[i]]` and `UsedByCurrPicLq[i]` is set equal to `used_by_curr_pic_lt_sps_flag[lt_idx_sps[i]]`.

Otherwise, `PocLsbLq[i]` is set equal to `poc_lsb_lt[i]` and `UsedByCurrPicLq[i]` is set equal to `used_by_curr_pic_lt_flag[i]`.

`delta_poc_msb_present_flag[i]` equal to 1 specifies that `delta_poc_msb_cycle_lt[i]` is present. `delta_poc_msb_present_flag[i]` equal to 0 specifies that `delta_poc_msb_cycle_lt[i]` is not present.

In some embodiments, let `prevTid0Pic` be the previous picture in decoding order that has `Temporand` equal to 0 and is not a RASL, RADL or SLNR picture. Let `setOfPrevPocVals` be a set consisting of the following:

the `PicOrderCntVal` of `prevTid0Pic`,
the `PicOrderCntVal` of each picture in the RPS of `prevTid0Pic`,

the `PicOrderCntVal` of each picture that follows `prevTid0Pic` in decoding order and precedes the current picture in decoding order.

In some embodiments, when there is more than one value in `setOfPrevPocVals` for which the value modulo `MaxPicOrderCntLsb` is equal to `PocLsbLt[i]`, `delta_poc_msb_present_flag[i]` shall be equal to 1.

`delta_poc_msb_cycle_lt[i]` is used to determine the value of the most significant bits of the picture order count value of the *i*-th entry in the long-term RPS of the current picture. When `delta_poc_msb_cycle_lt[i]` is not present, it is inferred to be equal to 0.

In some embodiments, the variable `DeltaPocMsbCycleLt[i]` is derived as follows:

if ($i=0$ || $i=\text{num_long_term_sps}$) `DeltaPocMsbCycleLt[i]` = `delta_poc_msb_cycle_lt[i]` else `DeltaPocMsbCycleLt[i]` = `delta_poc_msb_cycle_lt[i]+DeltaPocMsbCycleLt[i-1]`

1.2 Examples of Motion Vector Prediction (MVP) Between Short-Term and Long-Term Reference Pictures

In some embodiments, the motion vector prediction is only allowed if the target reference picture type and the predicted reference picture type is the same. In other words, when the types are different, motion vector prediction is disallowed.

Advanced Motion Vector Prediction (AMVP) is an example of motion vector prediction that includes an existing implementation. The relevant portion of the existing AMVP implementation is detailed below.

The motion vector `mvLXA` and the availability flag `availableFlagLXA` are derived in the following ordered steps:

(1) The sample location (`xNbA0`, `yNbA0`) is set equal to (`xPb-1`, `yPb+nPbH`) and the sample location (`xNbA1`, `yNbA1`) is set equal to (`xNbA0`, `yNbA0-1`).

(7) When `availableFlagLXA` is equal to 0, the following applies for (`xNbAk`, `yNbAk`) from (`xNbA0`, `yNbA0`) to (`xNbA1`, `yNbA1`) or until `availableFlagLXA` is equal to 1:

When `availableAk` is equal to TRUE and `availableFlagLXA` is equal to 0, the following applies:

If `PredFlagLX[xNbAk][yNbAk]` is equal to 1 and `LongTermRefPic(currPic, currPb, refIdxLX, RefPicListX)` is equal to `LongTermRefPic(currPic, currPb, refIdxLX[xNbAk][yNbAk], RefPicListX)`, `availableFlagLXA` is set equal to 1 and the following assignments are made:

$mvLXA = mvLX[xNbAk][yNbAk]$

$refIdxA = refIdxLX[xNbAk][yNbAk]$

$refPicListA = refPicListX$

Otherwise, when `PredFlagLY[xNbAk][yNbAk]` (with $Y=X$) is equal to 1 and `LongTermRefPic(currPic, currPb, refIdxLX, RefPicListX)` is equal to `LongTermRefPic(currPic, currPb, refIdxLY[xNbAk][yNbAk], RefPicListY)`, `availableFlagLXA` is set to 1.

The motion vector `mvLXB` and the availability flag `availableFlagLXB` are derived in the following ordered steps:

(1) The sample locations (`xNbB1`, `yNbB1`), (`xNbB1`, `yNbB1`) and (`xNbB2`, `yNbB2`) are set equal to (`xPb+nPbW`, `yPb-1`), (`xPb+nPbW-1`, `yPb-1`) and (`xPb-1`, `yPb-1`), respectively.

(5) When `isScaledFlagLX` is equal to 0, `availableFlagLXB` is set equal to 0 and the following applies for (`xNbBk`, `yNbBk`) from (`xNbB0`, `yNbB0`) to (`xNbB2`, `yNbB2`) or until `availableFlagLXB` is equal to 1:

The availability derivation process for a prediction block as specified in clause 6.4.2 is invoked with the luma location (xCb, yCb), the current luma coding block size nCbS, the luma location (xPb, yPb), the luma prediction block width nPbW, the luma prediction block height nPbH, the luma location (xNbY, yNbY) set equal to (xNbBk, yNbBk) and the partition index partIdx as inputs, and the output is assigned to the prediction block availability flag availableBk.

When availableBk is equal to TRUE and availableFlagLXB is equal to 0, the following applies:

If PredFlagLX[xNbBk][yNbBk] is equal to 1 and LongTermRefPic(currPic, currPb, refIdxLX, RefPicListX) is equal to LongTermRefPic(currPic, currPb, RefIdxLX[xNbBk][yNbBk], RefPicListX), availableFlagLXB is set equal to 1 and the following assignments are made:

$$mvLXB = MvLX[xNbBk][yNbBk]$$

$$refIdxB = RefIdxLX[xNbBk][yNbBk]$$

$$refPicListB = RefPicListX$$

Otherwise, when PredFlagLY[xNbBk][yNbBk] (with Y=!X) is equal to 1 and LongTermRefPic(currPic, currPb, refIdxLX, RefPicListX) is equal to LongTermRefPic(currPic, currPb, RefIdxLY[xNbBk][yNbBk], RefPicListY), availableFlagLXB is set equal to 1 and the following assignments are made:

$$mvLXB = MvLY[xNbBk][yNbBk]$$

Temporal Motion Vector Prediction (TMVP) is another example of motion vector prediction that includes an existing implementation. The relevant portion of the existing TMVP implementation is detailed below.

The variables mvLXCol and availableFlagLXCol are derived as follows:

If LongTermRefPic(currPic, currPb, refIdxLX, LX) is not equal to LongTermRefPic(ColPic, colPb, refIdxCol, listCol), both components of mvLXCol are set equal to 0 and availableFlagLXCol is set equal to 0.

Otherwise, the variable availableFlagLXCol is set equal to 1, refPicListCol[refIdxCol] is set to be the picture with reference index refIdxCol in the reference picture list listCol of the slice containing prediction block colPb in the collocated picture specified by ColPic.

2. Example Embodiments of Intra-Block Copy (IBC)

Intra-block copy (IBC) has been extends the concept of motion compensation from inter-frame coding to intra-frame coding. As shown in FIG. 1, the current block is predicted by a reference block in the same picture when IBC is applied. The samples in the reference block must have been already reconstructed before the current block is coded or decoded. Although IBC is relatively less efficient for most camera-captured sequences, it shows significant coding gains for screen content. The reason is that there are lots of reduplicated patterns, such as icons and text characters in a screen content picture. IBC can effectively remove the redundancy between these reduplicated patterns.

In HEVC-SCC, an inter-coded coding unit (CU) can apply IBC if it chooses the current picture as its reference picture. The motion vector (MV) is renamed as block vector (BV) in this case, and a BV always has an integer-pixel precision. To be compatible with main profile HEVC, the current picture is marked as a “long-term” reference picture

in the Decoded Picture Buffer (DPB). It should be noted that similarly, in multiple view/three-dimensional (3D) video coding standards, the inter-view reference picture is also marked as a “long-term” reference picture.

2.1 Embodiments of Picture Marking When IBC is Enabled

Semantics related to IBC in picture parameter set (PPS). pps_curr_pic_ref_enabled_flag equal to 1 specifies that a picture referring to the PPS may be included in a reference picture list of a slice of the picture itself. pps_curr_pic_ref_enabled_flag equal to 0 specifies that a picture referring to the PPS is never included in a reference picture list of a slice of the picture itself. When not present, the value of pps_curr_pic_ref_enabled_flag is inferred to be equal to 0.

It is a requirement of bitstream conformance that when sps_curr_pic_ref_enabled_flag is equal to 0, the value of pps_curr_pic_ref_enabled_flag shall be equal to 0.

The variable TwoVersionsOfCurrDecPicFlag is derived as follows:

$$\text{TwoVersionsOfCurrDecPicFlag} = \text{pps_curr_pic_ref_enabled_flag} \ \&\& \ (\text{sample_adaptive_offset_enabled_flag}! \text{pps_deblocking_filter_disabled_flag} | \text{deblocking_filter_override_enabled_flag})$$

When sps_max_dec_pic_buffering_minus1[TemporandId] is equal to 0, the value of TwoVersionsOfCurrDecPicFlag shall be equal to 0.

Decoding process. The current decoded picture after the invocation of the in-loop filter process is stored in the DPB in an empty picture storage buffer, the DPB fullness is incremented by one and this picture is marked as “used for short-term reference”.

When TwoVersionsOfCurrDecPicFlag is equal to 1, the current decoded picture before the invocation of the in-loop filter process as specified in clause F.8.7 [1] is stored in the DPB in an empty picture storage buffer, the DPB fullness is incremented by one, and this picture is marked as “used for long-term reference”.

3. Examples of the Joint Exploration Model (JEM)

In some embodiments, future video coding technologies are explored using a reference software known as the Joint Exploration Model (JEM). In JEM, sub-block based prediction is adopted in several coding tools, such as affine prediction, alternative temporal motion vector prediction (ATMVP), spatial-temporal motion vector prediction (STMVP), bi-directional optical flow (BIO), Frame-Rate Up Conversion (FRUC), Locally Adaptive Motion Vector Resolution (LAMVR), Overlapped Block Motion Compensation (OBMC), Local Illumination Compensation (LIC), and Decoder-side Motion Vector Refinement (DMVR).

3.1 Examples of Affine Prediction

In HEVC, only a translation motion model is applied for motion compensation prediction (MCP). However, the camera and objects may have many kinds of motion, e.g., zoom in/out, rotation, perspective motions, and/or other irregular motions. JEM, on the other hand, applies a simplified affine transform motion compensation prediction. FIG. 2 shows an example of an affine motion field of a block 200 described by two control point motion vectors V_0 and V_1 . The motion vector field (MVf) of the block 200 can be described by the following equation:

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$$\begin{cases} v_x = \frac{(v_{1x} - v_{0x})}{w}x - \frac{(v_{1y} - v_{0y})}{w}y + v_{0x} \\ v_y = \frac{(v_{1y} - v_{0y})}{w}x - \frac{(v_{1x} - v_{0x})}{w}y + v_{0y} \end{cases} \quad \text{Eq. (1)}$$

As shown in FIG. 2, (v_{0x}, v_{0y}) is motion vector of the top-left corner control point, and (v_{1x}, v_{1y}) is motion vector of the top-right corner control point. To simplify the motion compensation prediction, sub-block based affine transform prediction can be applied. The sub-block size $M \times N$ is derived as follows:

$$\begin{cases} M = \text{clip3}\left(4, w, \frac{w \times \text{MvPre}}{\max(\text{abs}(v_{1x} - v_{0x}), \text{abs}(v_{1y} - v_{0y}))}\right) \\ N = \text{clip3}\left(4, h, \frac{h \times \text{MvPre}}{\max(\text{abs}(v_{2x} - v_{0x}), \text{abs}(v_{2y} - v_{0y}))}\right) \end{cases} \quad \text{Eq. (2)}$$

Here, MvPre is the motion vector fraction accuracy (e.g., $1/16$ in JEM). (v_{2x}, v_{2y}) is motion vector of the bottom-left control point, calculated according to Eq. (1). M and N can be adjusted downward if necessary to make it a divisor of w and h, respectively.

FIG. 3 shows an example of affine MVF per sub-block for a block 300. To derive motion vector of each $M \times N$ sub-block, the motion vector of the center sample of each sub-block can be calculated according to Eq. (1), and rounded to the motion vector fraction accuracy (e.g., $1/16$ in JEM). Then the motion compensation interpolation filters can be applied to generate the prediction of each sub-block with derived motion vector. After the MCP, the high accuracy motion vector of each sub-block is rounded and saved as the same accuracy as the normal motion vector.

In the JEM, there are two affine motion modes: AF_INTER mode and AF_MERGE mode. For CUs with both width and height larger than 8, AF_INTER mode can be applied. An affine flag in CU level is signaled in the bitstream to indicate whether AF_INTER mode is used. In the AF_INTER mode, a candidate list with motion vector pair $\{(v_0, v_1), v_0 = \{V_A, V_B, V_C\}, v_1 = \{v_D, v_E\}\}$ is constructed using the neighboring blocks.

FIG. 4 shows an example of motion vector prediction (MVP) for a block 400 in the AF_INTER mode. As shown in FIG. 4, v_0 is selected from the motion vectors of the sub-block A, B, or C. The motion vectors from the neighboring blocks can be scaled according to the reference list. The motion vectors can also be scaled according to the relationship among the Picture Order Count (POC) of the reference for the neighboring block, the POC of the reference for the current CU, and the POC of the current CU. The approach to select v_1 from the neighboring sub-block D and E is similar. If the number of candidate list is smaller than 2, the list is padded by the motion vector pair composed by duplicating each of the AMVP candidates. When the candidate list is larger than 2, the candidates can first be sorted according to the neighboring motion vectors (e.g., based on the similarity of the two motion vectors in a pair candidate). In some implementations, the first two candidates are kept. In some embodiments, a Rate Distortion (RD) cost check is used to determine which motion vector pair candidate is selected as the control point motion vector prediction (CPMVP) of the current CU. An index indicating the position of the CPMVP in the candidate list can be signaled in the bitstream. After the CPMVP of the current affine CU is determined, affine motion estimation is applied and the

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control point motion vector (CPMV) is found. Then the difference of the CPMV and the CPMVP is signaled in the bitstream.

When a CU is applied in AF_MERGE mode, it gets the first block coded with an affine mode from the valid neighboring reconstructed blocks. FIG. 5A shows an example of the selection order of candidate blocks for a current CU 500. As shown in FIG. 5A, the selection order can be from left (501), above (502), above right (503), left bottom (504) to above left (505) of the current CU 500. FIG. 5B shows another example of candidate blocks for a current CU 500 in the AF_MERGE mode. If the neighboring left bottom block 501 is coded in affine mode, as shown in FIG. 5B, the motion vectors v_2 , v_3 and v_4 of the top left corner, above right corner, and left bottom corner of the CU containing the sub-block 501 are derived. The motion vector v_0 of the top left corner on the current CU 500 is calculated based on v_2 , v_3 and v_4 . The motion vector v_1 of the above right of the current CU can be calculated accordingly.

After the CPMV of the current CU v_0 and v_1 are computed according to the affine motion model in Eq. (1), the MVF of the current CU can be generated. In order to identify whether the current CU is coded with AF_MERGE mode, an affine flag can be signaled in the bitstream when there is at least one neighboring block is coded in affine mode.

3.2 Examples of Alternative Temporal Motion Vector Prediction (ATMVP)

In the ATMVP method, the temporal motion vector prediction (TMVP) method is modified by fetching multiple sets of motion information (including motion vectors and reference indices) from blocks smaller than the current CU.

FIG. 6 shows an example of ATMVP motion prediction process for a CU 600. The ATMVP method predicts the motion vectors of the sub-CUs 601 within a CU 600 in two steps. The first step is to identify the corresponding block 651 in a reference picture 650 with a temporal vector. The reference picture 650 is also referred to as the motion source picture. The second step is to split the current CU 600 into sub-CUs 601 and obtain the motion vectors as well as the reference indices of each sub-CU from the block corresponding to each sub-CU.

In the first step, a reference picture 650 and the corresponding block is determined by the motion information of the spatial neighboring blocks of the current CU 600. To avoid the repetitive scanning process of neighboring blocks, the first merge candidate in the merge candidate list of the current CU 600 is used. The first available motion vector as well as its associated reference index are set to be the temporal vector and the index to the motion source picture. This way, the corresponding block may be more accurately identified, compared with TMVP, in which the corresponding block (sometimes called collocated block) is always in a bottom-right or center position relative to the current CU.

In the second step, a corresponding block of the sub-CU 651 is identified by the temporal vector in the motion source picture 650, by adding to the coordinate of the current CU the temporal vector. For each sub-CU, the motion information of its corresponding block (e.g., the smallest motion grid that covers the center sample) is used to derive the motion information for the sub-CU. After the motion information of a corresponding $N \times N$ block is identified, it is converted to the motion vectors and reference indices of the current sub-CU, in the same way as TMVP of HEVC, wherein motion scaling and other procedures apply. For

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example, the decoder checks whether the low-delay condition (e.g., the POCs of all reference pictures of the current picture are smaller than the POC of the current picture) is fulfilled and possibly uses motion vector MV_x (e.g., the motion vector corresponding to reference picture list X) to predict motion vector MV_y (e.g., with X being equal to 0 or 1 and Y being equal to 1-X) for each sub-CU.

3.3 Examples of Spatial-Temporal Motion Vector Prediction (STMVP)

In the STMVP method, the motion vectors of the sub-CUs are derived recursively, following raster scan order. FIG. 7 shows an example of one CU with four sub-blocks and neighboring blocks. Consider an 8x8 CU **700** that includes four 4x4 sub-CUs A (**701**), B (**702**), C (**703**), and D (**704**). The neighboring 4x4 blocks in the current frame are labelled as a (**711**), b (**712**), c (**713**), and d (**714**).

The motion derivation for sub-CU A starts by identifying its two spatial neighbors. The first neighbor is the N×N block above sub-CU A **701** (block c **713**). If this block c (**713**) is not available or is intra coded the other N×N blocks above sub-CU A (**701**) are checked (from left to right, starting at block c **713**). The second neighbor is a block to the left of the sub-CU A **701** (block b **712**). If block b (**712**) is not available or is intra coded other blocks to the left of sub-CU A **701** are checked (from top to bottom, starting at block b **712**). The motion information obtained from the neighboring blocks for each list is scaled to the first reference frame for a given list. Next, temporal motion vector predictor (TMVP) of sub-block A **701** is derived by following the same procedure of TMVP derivation as specified in HEVC. The motion information of the collocated block at block D **704** is fetched and scaled accordingly. Finally, after retrieving and scaling the motion information, all available motion vectors are averaged separately for each reference list. The averaged motion vector is assigned as the motion vector of the current sub-CU.

3.4 Examples of Bi-Directional Optical Flow (BIO)

The bi-directional optical flow (BIO) method is a sample-wise motion refinement performed on top of block-wise motion compensation for bi-prediction. In some implementations, the sample-level motion refinement does not use signaling. Let $I^{(k)}$ be the luma value from reference k (k=0, 1) after block motion compensation, and $\partial I^{(k)}/\partial x$, $\partial I^{(k)}/\partial y$ are horizontal and vertical components of the $I^{(k)}$ gradient, respectively.

Assuming the optical flow is valid, the motion vector field (v_x, v_y) is given by:

$$\partial I^{(k)}/\partial t + v_x \partial I^{(k)}/\partial x + v_y \partial I^{(k)}/\partial y = 0 \quad \text{Eq. (3)}$$

Combining this optical flow equation with Hermite interpolation for the motion trajectory of each sample results in a unique third-order polynomial that matches both the function values $I^{(k)}$ and derivatives $\partial I^{(k)}/\partial x$, $\partial I^{(k)}/\partial y$ at the ends. The value of this polynomial at t=0 is the BIO prediction:

$$pred_{BIO} = \frac{1}{2} \cdot (I^{(0)} + I^{(1)} + v_x \cdot (\tau_1 \partial I^{(1)}/\partial x - \tau_0 \partial I^{(0)}/\partial x) + v_y \cdot (\tau_1 \partial I^{(1)}/\partial y - \tau_0 \partial I^{(0)}/\partial y)) \quad \text{Eq. (4)}$$

FIG. 8 shows an example optical flow trajectory in the Bi-directional Optical Flow (BIO) method. Here, τ_0 and τ_1 denote the distances to the reference frames. Distances τ_0 and τ_1 are calculated based on POC for Ref₀ and Ref₁: $\tau_0 = \text{POC}(\text{current}) - \text{POC}(\text{Ref}_0)$, $\tau_1 = \text{POC}(\text{Ref}_1) - \text{POC}(\text{current})$.

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If both predictions come from the same time direction (either both from the past or both from the future) then the signs are different (e.g., $\tau_0, \tau_1 < 0$). In this case, BIO is applied if the prediction is not from the same time moment (e.g., $\tau_0 \neq \tau_1$). Both referenced regions have non-zero motion (e.g., $MV_{x0}, MV_{y0}, MV_{x1}, MV_{y1} \neq 0$) and the block motion vectors are proportional to the time distance (e.g., $MV_{x0}/MV_{x1} = MV_{y0}/MV_{y1} = -\tau_0/\tau_1$).

The motion vector field (v_x, v_y) is determined by minimizing the difference Δ between values in points A and B. FIGS. 9A-9B show an example of intersection of motion trajectory and reference frame planes. Model uses only first linear term of a local Taylor expansion for Δ :

$$\Delta = (I^{(0)} - I^{(1)} + v_x (\tau_1 \partial I^{(1)}/\partial x + \tau_0 \partial I^{(0)}/\partial x) + v_y (\tau_1 \partial I^{(1)}/\partial y + \tau_0 \partial I^{(0)}/\partial y)) \quad \text{Eq. (5)}$$

All values in the above equation depend on the sample location, denoted as (i', j') . Assuming the motion is consistent in the local surrounding area, Δ can be minimized inside the $(2M+1) \times (2M+1)$ square window Ω centered on the currently predicted point (i, j) , where M is equal 2:

$$(v_x, v_y) = \underset{v_x, v_y}{\text{argmin}} \sum_{i', j' \in \Omega} \Delta^2[i', j'] \quad \text{Eq. (6)}$$

For this optimization problem, the JEM uses a simplified approach making first a minimization in the vertical direction and then in the horizontal direction. This results in the following:

$$v_x = (s_1 + r) > m ? \text{clip3} \left(-thBIO, thBIO, -\frac{s_3}{(s_1 + r)} \right) : 0 \quad \text{Eq. (7)}$$

$$v_y = (s_5 + r) > m ? \text{clip3} \left(-thBIO, thBIO, -\frac{s_6 - v_x s_2 / 2}{(s_5 + r)} \right) : 0 \quad \text{Eq. (8)}$$

$$\text{Where, } s_1 = \sum_{i', j' \in \Omega} (\tau_1 \partial I^{(1)} / \partial x + \tau_0 \partial I^{(0)} / \partial x)^2; \quad \text{Eq. (9)}$$

$$s_3 = \sum_{i', j' \in \Omega} (I^{(1)} - I^{(0)}) (\tau_1 \partial I^{(1)} / \partial x + \tau_0 \partial I^{(0)} / \partial x);$$

$$s_2 = \sum_{i', j' \in \Omega} (\tau_1 \partial I^{(1)} / \partial x + \tau_0 \partial I^{(0)} / \partial x) (\tau_1 \partial I^{(1)} / \partial y + \tau_0 \partial I^{(0)} / \partial y);$$

$$s_5 = \sum_{i', j' \in \Omega} (\tau_1 \partial I^{(1)} / \partial y + \tau_0 \partial I^{(0)} / \partial y)^2;$$

$$s_6 = \sum_{i', j' \in \Omega} (I^{(1)} - I^{(0)}) (\tau_1 \partial I^{(1)} / \partial y + \tau_0 \partial I^{(0)} / \partial y)$$

In order to avoid division by zero or a very small value, regularization parameters r and m can be introduced in Eq. (7) and Eq. (8), where:

$$r = 500 \cdot 4^{d-8} \quad \text{Eq. (10)}$$

$$m = 700 \cdot 4^{d-8} \quad \text{Eq. (11)}$$

Here, d is a bit depth of the video samples.

In order to keep the memory access for BIO the same as for regular bi-predictive motion compensation, all prediction and gradients values, $I^{(k)}$, $\partial I^{(k)}/\partial x$, $\partial I^{(k)}/\partial y$, are calculated for positions inside the current block. FIG. 9A shows an example of access positions outside of a block **900**. As shown in FIG. 9A, in Eq. (9), $(2M+1) \times (2M+1)$ square window Ω centered in currently predicted point on a boundary of predicted block needs to access positions outside of

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the block. In the JEM, values of $I^{(k)}$, $\partial I^{(k)}/\partial x$, $\partial I^{(k)}/\partial y$ outside of the block are set to be equal to the nearest available value inside the block. For example, this can be implemented as a padding area 901, as shown in FIG. 9B.

With BIO, it is possible that the motion field can be refined for each sample. To reduce the computational complexity, a block-based design of BIO is used in the JEM. The motion refinement can be calculated based on a 4x4 block. In the block-based BIO, the values of s_n in Eq. (9) of all samples in a 4x4 block can be aggregated, and then the aggregated values of s_n are used to derived BIO motion vectors offset for the 4x4 block. More specifically, the following formula can be used for block-based BIO derivation:

$$\begin{aligned}
 s_{1,b_k} &= \sum_{(x,y) \in b_k} \sum_{[l',j] \in \Omega(x,y)} (\tau_1 \partial I^{(1)} / \partial x + \tau_0 \partial I^{(0)} / \partial x)^2; \\
 s_{3,b_k} &= \sum_{(x,y) \in b_k} \sum_{[l',j] \in \Omega} (I^{(1)} - I^{(0)}) (\tau_1 \partial I^{(1)} / \partial x + \tau_0 \partial I^{(0)} / \partial x); \\
 s_{2,b_k} &= \sum_{(x,y) \in b_k} \sum_{[l',j] \in \Omega} (\tau_1 \partial I^{(1)} / \partial x + \tau_0 \partial I^{(0)} / \partial x) (\tau_1 \partial I^{(1)} / \partial y + \tau_0 \partial I^{(0)} / \partial y); \\
 s_{5,b_k} &= \sum_{(x,y) \in b_k} \sum_{[l',j] \in \Omega} (\tau_1 \partial I^{(1)} / \partial y + \tau_0 \partial I^{(0)} / \partial y)^2; \\
 s_{6,b_k} &= \sum_{(x,y) \in b_k} \sum_{[l',j] \in \Omega} (I^{(1)} - I^{(0)}) (\tau_1 \partial I^{(1)} / \partial y + \tau_0 \partial I^{(0)} / \partial y)
 \end{aligned}
 \tag{Eq. (12)}$$

Here, b_k denotes the set of samples belonging to the k-th 4x4 block of the predicted block. s_n in Eq (7) and Eq (8) are replaced by $((s_{n,b_k}) \gg 4)$ to derive the associated motion vector offsets.

In some scenarios, MV regiment of BIO may be unreliable due to noise or irregular motion. Therefore, in BIO, the magnitude of MV regiment is clipped to a threshold value. The threshold value is determined based on whether the reference pictures of the current picture are all from one direction. For example, if all the reference pictures of the current picture are from one direction, the value of the threshold is set to $12 \times 2^{14-d}$; otherwise, it is set to $12 \times 2^{13-d}$.

Gradients for BIO can be calculated at the same time with motion compensation interpolation using operations consistent with HEVC motion compensation process (e.g., two-dimensional (2D) separable Finite Impulse Response (FIR)). In some embodiments, the input for the 2D separable FIR is the same reference frame sample as for motion compensation process and fractional position (fracX, fracY) according to the fractional part of block motion vector. For horizontal gradient $\partial I / \partial x$, a signal is first interpolated vertically using BIOfilterS corresponding to the fractional position fracY with de-scaling shift d-8. Gradient filter BIOfilterG is then applied in horizontal direction corresponding to the fractional position fracX with de-scaling shift by 18-d. For vertical gradient $\partial I / \partial y$, a gradient filter is applied vertically using BIOfilterG corresponding to the fractional position fracY with de-scaling shift d-8. The signal displacement is then performed using BIOfilterS in horizontal direction corresponding to the fractional position fracX with de-scaling shift by 18-d. The length of interpolation filter for gradients calculation BIOfilterG and signal displacement BIOfilterF can be shorter (e.g., 6-tap) in order to maintain reasonable complexity. Table 2 shows example filters that can be used for gradients calculation of different fractional positions of block motion vector in BIO. Table 3 shows example interpolation filters that can be used for prediction signal generation in BIO.

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TABLE 2

Example filters for gradient calculation in BIO	
Fractional pel position	Interpolation filter for gradient(BIOfilterG)
0	{8, -39, -3, 46, -17, 5}
1/16	{8, -32, -13, 50, -18, 5}
1/8	{7, -27, -20, 54, -19, 5}
3/16	{6, -21, -29, 57, -18, 5}
1/4	{4, -17, -36, 60, -15, 4}
5/16	{3, -9, -44, 61, -15, 4}
3/8	{1, -4, -48, 61, -13, 3}
7/16	{0, 1, -54, 60, -9, 2}
1/2	{-1, 4, -57, 57, -4, 1}

TABLE 3

Example interpolation filters for prediction signal generation in BIO	
Fractional pel position	Interpolation filter for prediction signal(BIOfilterS)
0	{0, 0, 64, 0, 0, 0}
1/16	{1, -3, 64, 4, -2, 0}
1/8	{1, -6, 62, 9, -3, 1}
3/16	{2, -8, 60, 14, -5, 1}
1/4	{2, -9, 57, 19, -7, 2}
5/16	{3, -10, 53, 24, -8, 2}
3/8	{3, -11, 50, 29, -9, 2}
7/16	{3, -11, 44, 35, -10, 3}
1/2	{3, -10, 35, 44, -11, 3}

In the JEM, BIO can be applied to all bi-predicted blocks when the two predictions are from different reference pictures. When Local Illumination Compensation (LIC) is enabled for a CU, BIO can be disabled.

In some embodiments, OBMC is applied for a block after normal MC process. To reduce the computational complexity, BIO may not be applied during the OBMC process. This means that BIO is applied in the MC process for a block when using its own MV and is not applied in the MC process when the MV of a neighboring block is used during the OBMC process.

3.5 Examples of Frame-rate Up Conversion (FRUC)

A FRUC flag can be signaled for a CU when its merge flag is true. When the FRUC flag is false, a merge index can be signaled and the regular merge mode is used. When the FRUC flag is true, an additional FRUC mode flag can be signaled to indicate which method (e.g., bilateral matching or template matching) is to be used to derive motion information for the block.

At the encoder side, the decision on whether using FRUC merge mode for a CU is based on RD cost selection as done for normal merge candidate. For example, multiple matching modes (e.g., bilateral matching and template matching) are checked for a CU by using RD cost selection. The one leading to the minimal cost is further compared to other CU modes. If a FRUC matching mode is the most efficient one, FRUC flag is set to true for the CU and the related matching mode is used.

Typically, motion derivation process in FRUC merge mode has two steps: a CU-level motion search is first performed, then followed by a Sub-CU level motion refinement. At CU level, an initial motion vector is derived for the whole CU based on bilateral matching or template matching.

First, a list of MV candidates is generated and the candidate that leads to the minimum matching cost is selected as the starting point for further CU level refinement. Then a local search based on bilateral matching or template matching around the starting point is performed. The MV results in the minimum matching cost is taken as the MV for the whole CU. Subsequently, the motion information is further refined at sub-CU level with the derived CU motion vectors as the starting points.

For example, the following derivation process is performed for a $W \times H$ CU motion information derivation. At the first stage, MV for the whole $W \times H$ CU is derived. At the second stage, the CU is further split into $M \times M$ sub-CUs. The value of M is calculated as in (16), D is a predefined splitting depth which is set to 3 by default in the JEM. Then the MV for each sub-CU is derived.

$$M = \max\left\{4, \min\left\{\frac{M}{2^D}, \frac{N}{2^D}\right\}\right\} \quad \text{Eq. (13)}$$

FIG. 10 shows an example of bilateral matching used in the Frame-Rate Up Conversion (FRUC) method. The bilateral matching is used to derive motion information of the current CU by finding the closest match between two blocks along the motion trajectory of the current CU (1000) in two different reference pictures (1010, 1011). Under the assumption of continuous motion trajectory, the motion vectors MV0 (1001) and MV1 (1002) pointing to the two reference blocks are proportional to the temporal distances, e.g., TD0 (1003) and TD1 (1004), between the current picture and the two reference pictures. In some embodiments, when the current picture 1000 is temporally between the two reference pictures (1010, 1011) and the temporal distance from the current picture to the two reference pictures is the same, the bilateral matching becomes minor based bi-directional MV.

FIG. 11 shows an example of template matching used in the Frame-Rate Up Conversion (FRUC) method. Template matching can be used to derive motion information of the current CU 1100 by finding the closest match between a template (e.g., top and/or left neighboring blocks of the current CU) in the current picture and a block (e.g., same size to the template) in a reference picture 1110. Except the aforementioned FRUC merge mode, the template matching can also be applied to AMVP mode. In both JEM and HEVC, AMVP has two candidates. With the template matching method, a new candidate can be derived. If the newly derived candidate by template matching is different to the first existing AMVP candidate, it is inserted at the very beginning of the AMVP candidate list and then the list size is set to two (e.g., by removing the second existing AMVP candidate). When applied to AMVP mode, only CU level search is applied.

The MV candidate set at CU level can include the following: (1) original AMVP candidates if the current CU is in AMVP mode, (2) all merge candidates, (3) several MVs in the interpolated MV field (described later), and top and left neighboring motion vectors.

When using bilateral matching, each valid MV of a merge candidate can be used as an input to generate a MV pair with the assumption of bilateral matching. For example, one valid MV of a merge candidate is (MV_a , ref_a) at reference list A. Then the reference picture ref_b of its paired bilateral MV is found in the other reference list B so that ref_a and ref_b are temporally at different sides of the current picture. If such a ref_b is not available in reference list B, ref_b is determined as

a reference which is different from ref_a and its temporal distance to the current picture is the minimal one in list B. After ref_b is determined, MV_b is derived by scaling MV_a based on the temporal distance between the current picture and ref_a , ref_b .

In some implementations, four MVs from the interpolated MV field can also be added to the CU level candidate list. More specifically, the interpolated MVs at the position (0, 0), ($W/2$, 0), (0, $H/2$) and ($W/2$, $H/2$) of the current CU are added. When FRUC is applied in AMVP mode, the original AMVP candidates are also added to CU level MV candidate set. In some implementations, at the CU level, 15 MVs for AMVP CUs and 13 MVs for merge CUs can be added to the candidate list.

The MV candidate set at sub-CU level includes an MV determined from a CU-level search, (2) top, left, top-left and top-right neighboring MVs, (3) scaled versions of collocated MVs from reference pictures, (4) one or more ATMVP candidates (e.g., up to four), and (5) one or more STMVP candidates (e.g., up to four). The scaled MVs from reference pictures are derived as follows. The reference pictures in both lists are traversed. The MVs at a collocated position of the sub-CU in a reference picture are scaled to the reference of the starting CU-level MV. ATMVP and STMVP candidates can be the four first ones. At the sub-CU level, one or more MVs (e.g., up to 17) are added to the candidate list.

Generation of an interpolated MV field. Before coding a frame, interpolated motion field is generated for the whole picture based on unilateral ME. Then the motion field may be used later as CU level or sub-CU level MV candidates.

In some embodiments, the motion field of each reference pictures in both reference lists is traversed at 4×4 block level. FIG. 12 shows an example of unilateral Motion Estimation (ME) 1200 in the FRUC method. For each 4×4 block, if the motion associated to the block passing through a 4×4 block in the current picture and the block has not been assigned any interpolated motion, the motion of the reference block is scaled to the current picture according to the temporal distance TD0 and TD1 (the same way as that of MV scaling of TMVP in HEVC) and the scaled motion is assigned to the block in the current frame. If no scaled MV is assigned to a 4×4 block, the block's motion is marked as unavailable in the interpolated motion field.

Interpolation and matching cost. When a motion vector points to a fractional sample position, motion compensated interpolation is needed. To reduce complexity, bi-linear interpolation instead of regular 8-tap HEVC interpolation can be used for both bilateral matching and template matching.

The calculation of matching cost is a bit different at different steps. When selecting the candidate from the candidate set at the CU level, the matching cost can be the absolute sum difference (SAD) of bilateral matching or template matching. After the starting MV is determined, the matching cost C of bilateral matching at sub-CU level search is calculated as follows:

$$C = SAD + w \cdot (|MV_x - MV_x^s| + |MV_y - MV_y^s|) \quad \text{Eq. (14)}$$

Here, w is a weighting factor. In some embodiments, w can be empirically set to 4. MV and MV^s indicate the current MV and the starting MV, respectively. SAD may still be used as the matching cost of template matching at sub-CU level search.

In FRUC mode, MV is derived by using luma samples only. The derived motion will be used for both luma and chroma for MC inter prediction. After MV is decided, final

MC is performed using 8-taps interpolation filter for luma and 4-taps interpolation filter for chroma.

MV refinement is a pattern based MV search with the criterion of bilateral matching cost or template matching cost. In the JEM, two search patterns are supported—an unrestricted center-biased diamond search (UCBDS) and an adaptive cross search for MV refinement at the CU level and sub-CU level, respectively. For both CU and sub-CU level MV refinement, the MV is directly searched at quarter luma sample MV accuracy, and this is followed by one-eighth luma sample MV refinement. The search range of MV refinement for the CU and sub-CU step are set equal to 8 luma samples.

In the bilateral matching merge mode, bi-prediction is applied because the motion information of a CU is derived based on the closest match between two blocks along the motion trajectory of the current CU in two different reference pictures. In the template matching merge mode, the encoder can choose among uni-prediction from list0, uni-prediction from list1, or bi-prediction for a CU. The selection can be based on a template matching cost as follows:

```

If costBi <= factor * min (cost0, cost1)
    bi-prediction is used;
Otherwise, if cost0 <= cost1
    uni-prediction from list0 is used;
Otherwise,
    uni-prediction from list1 is used;

```

Here, cost0 is the SAD of list0 template matching, cost1 is the SAD of list1 template matching and costBi is the SAD of bi-prediction template matching. For example, when the value of factor is equal to 1.25, it means that the selection process is biased toward bi-prediction. The inter prediction direction selection can be applied to the CU-level template matching process.

3.6 Examples of Locally Adaptive Motion Vector Resolution (LAMVR)

In HEVC, motion vector differences (MVDs) (between the motion vector and predicted motion vector of a PU) are signaled in units of quarter luma samples when use_integer_mv_flag is equal to 0 in the slice header. In the JEM, a locally adaptive motion vector resolution (LAMVR) is introduced. In the JEM, MVD can be coded in units of quarter luma samples, integer luma samples or four luma samples. The MVD resolution is controlled at the coding unit (CU) level, and MVD resolution flags are conditionally signaled for each CU that has at least one non-zero MVD components.

For a CU that has at least one non-zero MVD components, a first flag is signaled to indicate whether quarter luma sample MV precision is used in the CU. When the first flag (equal to 1) indicates that quarter luma sample MV precision is not used, another flag is signaled to indicate whether integer luma sample MV precision or four luma sample MV precision is used.

When the first MVD resolution flag of a CU is zero, or not coded for a CU (meaning all MVDs in the CU are zero), the quarter luma sample MV resolution is used for the CU. When a CU uses integer-luma sample MV precision or four-luma-sample MV precision, the MVPs in the AMVP candidate list for the CU are rounded to the corresponding precision.

On the encoder side, CU-level RD checks are used to determine which MVD resolution is to be used for a CU. That is, the CU-level RD check is performed three times for each MVD resolution. To accelerate encoder speed, the following encoding schemes are applied in the JEM:

During RD check of a CU with normal quarter luma sample MVD resolution, the motion information of the current CU (integer luma sample accuracy) is stored. The stored motion information (after rounding) is used as the starting point for further small range motion vector refinement during the RD check for the same CU with integer luma sample and 4 luma sample MVD resolution so that the time-consuming motion estimation process is not duplicated three times.

RD check of a CU with 4 luma sample MVD resolution is conditionally invoked. For a CU, when RD cost integer luma sample MVD resolution is much larger than that of quarter luma sample MVD resolution, the RD check of 4 luma sample MVD resolution for the CU is skipped.

3.7 Examples of Overlapped Block Motion Compensation (OBMC)

In the JEM, OBMC can be switched on and off using syntax at the CU level. When OBMC is used in the JEM, the OBMC is performed for all motion compensation (MC) block boundaries except the right and bottom boundaries of a CU. Moreover, it is applied for both the luma and chroma components. In the JEM, an MC block corresponds to a coding block. When a CU is coded with sub-CU mode (includes sub-CU merge, affine and FRUC mode), each sub-block of the CU is a MC block. To process CU boundaries in a uniform fashion, OBMC is performed at sub-block level for all MC block boundaries, where sub-block size is set equal to 4x4, as shown in FIGS. 13A and 13B.

FIG. 13A shows sub-blocks at the CU/PU boundary, and the hatched sub-blocks are where OBMC applies. Similarly, FIG. 13B shows the sub-PUs in ATMVP mode.

When OBMC applies to the current sub-block, besides current motion vectors, motion vectors of four connected neighboring sub-blocks, if available and are not identical to the current motion vector, are also used to derive prediction block for the current sub-block. These multiple prediction blocks based on multiple motion vectors are combined to generate the final prediction signal of the current sub-block.

Prediction block based on motion vectors of a neighboring sub-block is denoted as PN, with N indicating an index for the neighboring above, below, left and right sub-blocks and prediction block based on motion vectors of the current sub-block is denoted as PC. When PN is based on the motion information of a neighboring sub-block that contains the same motion information to the current sub-block, the OBMC is not performed from PN. Otherwise, every sample of PN is added to the same sample in PC, i.e., four rows/columns of PN are added to PC. The weighting factors $\{1/4, 1/8, 1/16, 1/32\}$ are used for PN and the weighting factors $\{3/4, 7/8, 15/16, 31/32\}$ are used for PC. The exception are small MC blocks, (i.e., when height or width of the coding block is equal to 4 or a CU is coded with sub-CU mode), for which only two rows/columns of PN are added to PC. In this case weighting factors $\{1/4, 1/8\}$ are used for PN and weighting factors $\{3/4, 7/8\}$ are used for PC. For PN generated based on motion vectors of vertically (horizontally) neighboring sub-block, samples in the same row (column) of PN are added to PC with a same weighting factor.

In the JEM, for a CU with size less than or equal to 256 luma samples, a CU level flag is signaled to indicate whether OBMC is applied or not for the current CU. For the CUs with size larger than 256 luma samples or not coded with AMVP mode, OBMC is applied by default. At the encoder, when OBMC is applied for a CU, its impact is taken into account during the motion estimation stage. The prediction signal formed by OBMC using motion information of the top neighboring block and the left neighboring block is used to compensate the top and left boundaries of the original signal of the current CU, and then the normal motion estimation process is applied.

3.8 Examples of Local Illumination Compensation (LIC)

LIC is based on a linear model for illumination changes, using a scaling factor a and an offset b . And it is enabled or disabled adaptively for each inter-mode coded coding unit (CU).

When LIC applies for a CU, a least square error method is employed to derive the parameters a and b by using the neighboring samples of the current CU and their corresponding reference samples. FIG. 14 shows an example of neighboring samples used to derive parameters of the IC algorithm. Specifically, and as shown in FIG. 14, the subsampled (2:1 subsampling) neighboring samples of the CU and the corresponding samples (identified by motion information of the current CU or sub-CU) in the reference picture are used. The IC parameters are derived and applied for each prediction direction separately.

When a CU is coded with merge mode, the LIC flag is copied from neighboring blocks, in a way similar to motion information copy in merge mode; otherwise, an LIC flag is signaled for the CU to indicate whether LIC applies or not.

When LIC is enabled for a picture, an additional CU level RD check is needed to determine whether LIC is applied or not for a CU. When LIC is enabled for a CU, the mean-removed sum of absolute difference (MR-SAD) and mean-removed sum of absolute Hadamard-transformed difference (MR-SATD) are used, instead of sum of absolute differences (SAD) and sum of absolute Hadamard-transformed difference (SATD), for integer pel motion search and fractional pel motion search, respectively.

To reduce the encoding complexity, the following encoding scheme is applied in the JEM:

LIC is disabled for the entire picture when there is no obvious illumination change between a current picture and its reference pictures. To identify this situation, histograms of a current picture and every reference picture of the current picture are calculated at the encoder. If the histogram difference between the current picture and every reference picture of the current picture is smaller than a given threshold, LIC is disabled for the current picture; otherwise, LIC is enabled for the current picture.

3.9 Examples of Decoder-side Motion Vector Refinement (DMVR)

In a bi-prediction operation, for the prediction of one block region, two prediction blocks, formed using a motion vector (MV) of list0 and a MV of list1, respectively, are combined to form a single prediction signal. In the decoder-side motion vector refinement (DMVR) method, the two motion vectors of the bi-prediction are further refined by a bilateral template matching process. The bilateral template

matching applied in the decoder to perform a distortion-based search between a bilateral template and the reconstruction samples in the reference pictures in order to obtain a refined MV without transmission of additional motion information.

In DMVR, a bilateral template is generated as the weighted combination (i.e., average) of the two prediction blocks, from the initial MV0 of list0 and MV1 of list1, respectively, as shown in FIG. 15. The template matching operation consists of calculating cost measures between the generated template and the sample region (around the initial prediction block) in the reference picture. For each of the two reference pictures, the MV that yields the minimum template cost is considered as the updated MV of that list to replace the original one. In the JEM, nine MV candidates are searched for each list. The nine MV candidates include the original MV and 8 surrounding MVs with one luma sample offset to the original MV in either the horizontal or vertical direction, or both. Finally, the two new MVs, i.e., MV0' and MV1' as shown in FIG. 15, are used for generating the final bi-prediction results. A sum of absolute differences (SAD) is used as the cost measure.

DMVR is applied for the merge mode of bi-prediction with one MV from a reference picture in the past and another from a reference picture in the future, without the transmission of additional syntax elements. In the JEM, when LIC, affine motion, FRUC, or sub-CU merge candidate is enabled for a CU, DMVR is not applied.

4. Exemplary Methods for IBC in Video Coding

FIG. 16 shows a flowchart of an exemplary method for video encoding using intra-block copy. The method 1600 includes, at step 1610, determining whether a current block of the current picture is to be encoded using a motion compensation algorithm. The method 1600 includes, in step 1620, encoding, based on the determining, the current block by selectively applying an intra-block copy to the current block. More generally, whether or not to apply the intra-block copy to the current block is based on whether the current block is to be encoded using a specific motion compensation algorithm.

FIG. 17 shows a flowchart of another exemplary method video encoding using intra-block copy. The method 1700 includes, at step 1710, determining whether a current block of the current picture is to be encoded using an intra-block copy. The method 1700 includes, in step 1720, encoding, based on the determining, the current block by selectively applying a motion compensation algorithm to the current block. More generally, whether or not to encode the current block using the motion compensation algorithm is based on whether the current block is to be encoded using the intra-block copy.

FIG. 18 shows a flowchart of an exemplary method for video decoding using intra-block copy. The method 1800 includes, at step 1810, determining whether a current block of the current picture is to be decoded using a motion compensation algorithm. The method 1800 includes, in step 1820, decoding, based on the determining, the current block by selectively applying an intra-block copy to the current block. More generally, whether or not to apply the intra-block copy to the current block is based on whether the current block is to be decoded using a specific motion compensation algorithm.

FIG. 19 shows a flowchart of another exemplary method video decoding using intra-block copy. The method 1900 includes, at step 1910, determining whether a current block

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of the current picture is to be decoded using an intra-block copy. The method 1900 includes, in step 1920, decoding, based on the determining, the current block by selectively applying a motion compensation algorithm to the current block. More generally, whether or not to decode the current block using the motion compensation algorithm is based on whether the current block is to be decoded using the intra-block copy.

The methods 1600, 1700, 1800 and 1900, described in the context of FIGS. 16-19, may further include are further the step of determining whether the motion compensation algorithm is compatible with the intra-block copy. The compatibility of the intra-block copy and the motion compensation algorithms are elucidated in the following examples described for different specific motion compensation algorithms.

Example 1

After determining that the current picture must be stored in the buffer for a duration corresponding to the duration of a long-term reference picture, the current picture is not marked as a “long-term” reference picture in a buffer (e.g., a decoded picture buffer). Instead, it is marked as a new type of reference picture different from “short-term” or “long-term”. For example, it may be marked as a “current” reference picture, a “self” reference picture, an “instant” reference picture, an “intra block copy” reference picture, and so on.

- (a) Similarly, prediction (e.g., motion information prediction) between pictures with the new picture type and short-term pictures, and/or prediction (e.g., motion information prediction) between pictures with the new picture type and long-term pictures are disallowed.
- (b) When the prediction between pictures with the new picture type and short/long-term pictures is allowed, scaling process of motion vectors may be skipped.

Example 2

It is proposed that affine prediction cannot be applied for IBC coded blocks. In this case, when a block is coded with IBC mode, the signaling of indications of affine prediction is skipped.

- (a) In an example, if at least one reference picture of the current block in a reference picture list is the current picture, the flag indicating whether affine prediction is used (a.k.a., `affine_flag`) is not signaled and inferred to be 0.
- (b) Alternatively, if the two reference pictures for the two reference lists are both the current picture, `affine_flag` is not signaled and inferred to be 0.
- (c) Alternatively, if a block is associated with M sets of motion information ($M > 2$), and all the associated reference pictures are the current picture, `affine_flag` is not signaled and inferred to be 0.
- (d) Alternatively, if reference pictures of the current block includes both the current picture and other reference pictures, `affine_flag` may be signaled. In this case, `affine_flag` only controls whether the affine prediction is used for the non-IBC inter-prediction, i.e., inter-prediction which is not from the current picture. Some examples of this case are as following:
 - (i) In an example, the current block is associated with two reference pictures, and only one reference picture is the current picture.

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- (ii) In an example, the current block is associated with more than two reference pictures from, and at least one of the reference pictures is the current picture and at least one of the reference pictures is not the current picture.
- (iii) Even if `affine_flag` is equal to 1, affine prediction is not used for the IBC prediction with the reference picture identical to the current picture.
- (iv) Alternatively, furthermore, even if `affine_flag` is equal to 1, affine prediction is not used for the reference list wherein the IBC prediction is enabled with the reference picture identical to the current picture.
- (e) In an example, whether and how to apply affine prediction for IBC coded blocks can be transmitted from the encoder to the decoder at sequence level, picture level, slice level, Coding Tree Unit (CTU) a.k.a., Largest Coding Unit (LCU) level, region level, CU level, or PU level. The information can be signaled in Sequence Parameter Set (SPS), Picture Parameter Set (PPS), Slice Header (SH), CTU (a.k.a., LCU), region, CU or PU.

Example 3

It is proposed that IBC cannot be applied for a block with affine prediction. In this case, when a block is coded with affine mode, the signaling of indications of IBC is skipped.

- (a) In an example, if `affine_flag` is 1 for a block, any reference picture of the current block cannot be the current picture. In another example, if `affine_flag` is 1 for a block, at least one reference picture of the current block is not identical to the current picture.
- (b) Alternatively, whether and how to apply IBC for a block with affine prediction can be transmitted from the encoder to the decoder at sequence level, picture level, slice level, Coding Tree Unit (CTU) (e.g., Largest Coding Unit (LCU) level, region level, CU level, or PU level). The information can be signaled in Sequence Parameter Set (SPS), Picture Parameter Set (PPS), Slice Header (SH), CTU (a.k.a., LCU), region, CU or PU.

Example 4

Alternatively, it is proposed that affine prediction can be applied for IBC coded blocks. In this case, the indications of affine and IBC may be both signaled. When affine prediction is applied for a IBC coded block, the following may further apply:

- (a) In an example, the MV Prediction (MVP) for control points (such as $\vec{v}_0$ and $\vec{v}_1$) can only be derived from previously decoded MVs with the current picture as the reference picture. MV scaling is disabled for MVP derivation. Vice versa, MVP for control points of a block with an affine prediction but not IBC-coded, can only be derived from previously decoded MVs with reference pictures not identical to the current picture.
- (b) In an example, MVs at control points explained in “Examples of affine prediction” section are signaled in the integer precision. MVPs for them should be truncated or rounded to the integer-pixel precision.
- (c) In an example, if MVs at control points (such as $\vec{v}_0$ and $\vec{v}_1$) are derived from neighboring MVs, such as in the

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affine-merge mode, the derived MVs are truncated or rounded to the integer-pixel precision.

- (d) In an example, a MV inside the block derived from MVs at the control points, such as (v_x, v_y) in eq. (1) are truncated or rounded to the integer-pixel precision.

- (e) In an example, OBMC is not applied.

- (f) In an example, the y component of the MV at control points (e.g., IN) is not signaled and is always considered as zero.

Example 5

It is proposed that the temporal vector used in the first step of ATMVP cannot be derived from a neighboring block coded with IBC. In one example, a neighboring block with the current picture as its reference picture is marked as “unavailable” or intra-coded in the first step of ATMVP.

Example 6

It is proposed that a sub-CU's corresponding block is marked as “unavailable” or intra-coded if the corresponding block is coded with IBC in the second step of ATMVP.

Example 7

Alternatively, it is proposed that the motion information of a sub-CU's corresponding block is copied to the Sub-CU without any scaling if corresponding block is coded with IBC in the second step of ATMVP. The Sub-CU applies IBC with the same MV as the corresponding block but the reference picture is changed to the current picture.

Example 8

More than one ATMVP candidates may be added wherein one may be derived from temporal neighboring blocks using above methods and the other one is derived from temporal neighboring blocks with at least one sub-CU with different way of deriving sub-CU motion information if the co-located sub-CU is coded with IBC.

Example 9

It is proposed a neighboring block is marked as “unavailable” or intra-coded if the corresponding block is coded with IBC in STMVP.

Example 10

Alternatively, it is proposed the motion information of a neighboring block is copied to the Sub-CU without any scaling if the neighboring block is coded with IBC in STMVP. The Sub-CU applies IBC with the same MV as the neighboring block.

- (a) In an example, there is no need to access the temporal motion information from a co-located temporal neighboring block in this case.

- (b) When there is more than one neighboring block of a sub-CU coded with IBC, those associated motion vectors may be jointly used to derive the motion information of the current sub-CU.

Example 11

It is proposed that the averaged (or other kinds of derivation function, like weighted average) motion vector can

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only be derived from MVs all referring to the current picture or all referring to a reference picture not identical to the current picture.

Example 12

More than one STMVP candidates may be added wherein one may be derived from neighboring blocks with non-current pictures of all sub-CUs and one may be derived from neighboring blocks with current-pictures of all sub-CUs.

Example 13

It is proposed that BIO cannot be applied for IBC coded blocks. In one example, if at least one reference picture of the current block is the current picture, BIO is not conducted in the current block.

Example 14

It is proposed that bilateral matching in FRUC is not valid if there is at least one reference picture identical to the current picture in the reference list. In this case, the FRUC mode flag is not signaled when the FRUC flag is 1. The FRUC mode is always inferred as template matching.

Example 15

It is proposed that template matching in FRUC only searches integer-pixels without searching the sub-pixels if the reference picture is the current picture.

Example 16

It is proposed that LAMVR cannot be applied for IBC coded blocks.

- (a) In an example, if at least one reference picture of the current block is the current picture, the information indicating the resolution of the MV is not signaled and inferred to be as default (such as $\frac{1}{4}$ pixel for non-IBC inter MC, integer pixel for IBC).

- (b) In another example, if the two reference pictures for the two reference lists are both the current picture, the information indicating the resolution of the MV is not signaled and inferred to be as default (such as integer pixel for IBC).

- (c) Alternatively, if reference pictures of the current block includes both the current picture and other reference pictures, the information indicating the resolution of the MV may be signaled, but it only controls the resolution of the MV of the non-IBC inter-prediction with the reference picture not identical to the current picture. The MV resolution for IBC prediction with the reference picture identical to the current picture is always integer pixel. Some examples of this case are as following:

- (i) In an example, the current block is associated with two reference pictures, and only one reference picture is the current picture.

- (ii) In an example, the current block is associated with more than two reference pictures from, and at least one of the reference pictures is the current picture and at least one of the reference pictures is not the current picture.

- (d) In another example, the MV resolution for IBC prediction with the reference picture identical to the current picture can be adaptive at block level and the

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information of MV resolution for IBC prediction can be signaled. But the possible resolutions are all larger than the integer-pixel, such as two-pixel, four-pixel and so on.

- (i) In a further example, whether the signaled MV resolution is applied on the IBC prediction depends on the MV resolution. For example, if the signaled MV resolution is smaller than integer-pixel such as $\frac{1}{4}$ pixel, it is not applied on the IBC prediction. If the signaled MV resolution is equal or larger than the integer-pixel, such as two-pixel, it is applied on the IBC prediction.
- (e) In an example, whether and how to apply LAMVR for IBC coded blocks can be transmitted from the encoder to the decoder at sequence level, picture level, slice level, Coding Tree Unit (CTU) (e.g., Largest Coding Unit (LCU) level, CU level, or PU level). The information can be signaled in Sequence Parameter Set (SPS), Picture Parameter Set (PPS), Slice Header (SH), CTU (a.k.a., LCU), CU or PU.

Example 17

It is proposed that OBMC cannot be applied for IBC coded blocks.

- (a) In an example, if at least one reference picture of the current block is the current picture, OBMC is not conducted in the current block.
- (b) In another example, if the reference picture of the current block in one reference list is the current picture, and the reference picture of the current block in another reference list is not the current picture, OBMC is only conducted on the inter-prediction from the latter reference list.

Example 18

It is proposed that LIC cannot be applied for IBC coded blocks.

- (a) In an example, if at least one reference picture of the current block is the current picture, the LIC flag is not signaled and inferred to be 0.
- (b) In another example, if the two reference pictures for the two reference lists are both the current picture, LIC flag is not signaled and inferred to be 0.
- (c) Alternatively, if reference pictures of the current block includes both the current picture and other reference pictures, LIC flag may be signaled, but it only controls whether LIC is used for the non-IBC inter-prediction with the reference picture not identical to the current picture. The LIC flag for IBC prediction with the reference picture identical to the current picture is always 0. Some examples of this case are as following:
- (i) In an example, the current block is associated with two reference pictures, and only one reference picture is the current picture.
- (ii) In an example, the current block is associated with more than two reference pictures from, and at least one of the reference pictures is the current picture and at least one of the reference pictures is not the current picture.
- (d) In an example, whether and how to apply LIC for IBC coded blocks can be transmitted from the encoder to the decoder at sequence level, picture level, slice level, Coding Tree Unit (CTU) (e.g., Largest Coding Unit (LCU) level, CU level, or PU level). The information

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can be signaled in Sequence Parameter Set (SPS), Picture Parameter Set (PPS), Slice Header (SH), CTU (a.k.a., LCU), CU or PU.

Example 19

It is proposed that DMVR cannot be applied for IBC coded blocks. In one example, if at least one reference picture of the current block is the current picture, DMVR is not conducted in the current block.

Listed below are some examples of the embodiments described in this application. A block, as used in this application, can be a contiguous or a noncontiguous collection of pixels, voxels, sub-pixels, and/or sub-voxels. For example, a block can be rectilinear, such as a 4×4 square, 6×4 rectangle, or curvilinear, such as an ellipse.

A portion of the visual information, as used in this application, can be a subset of visual information. A coded representation, as used in this application, can be a bitstream representing the visual information that has been encoded using one of the embodiments described in this application. An indicator, as used in this application, can be a flag or a field in the coded representation or can be multiple separate flags or fields.

A decoding technique, as used in this application can be applied by a decoder and can be implemented in hardware or software. The decoding technique can undo in reverse sequence everything a coder does. When an appropriate decoding technique is applied to an encoded representation, a visual information can be obtained as a result.

An initial block in the plurality of blocks, as used in this application, is a block occurring before the first block in the coded representation. A predetermined component, as used in this application, can be an x-axis, a y-axis or a z-axis component of a motion vector.

Additional Examples

1. As depicted in FIG. 23, A visual information decoding method (2300), comprising: determining (2302) that a block being decoded representing a portion of the visual information is coded using a first coding technique; and decoding (2304) the coded representation by using a first decoding technique corresponding to the first coding technique and by excluding use of a second decoding technique corresponding to a second coding technique; wherein one of the first and second coding techniques corresponds to an intra-block copy (IBC) technique that uses a second block of a same video picture for coding the block being decoded and the other corresponds to an affine coding technique that uses an affine motion model for coding the block being decoded.

2. As depicted in FIG. 22, a visual information processing method (2200) comprising: determining (2202) that a block being decoded representing a portion of an encoded picture of visual information is coded using an intra-block copy (IBC) technique that uses a second block of the picture for coding the block being decoded and an affine coding technique that uses an affine motion model for coding the block being decoded; and decoding (2204) the coded representation by using an IBC decoding technique corresponding to the IBC decoding technique and an affine decoding technique corresponding to the affine coding technique.

3. The method of examples 1-2, the IBC decoding technique comprising: obtaining a plurality of blocks representing the encoded picture and an indicator of use of the IBC technique; decoding a plurality of initial blocks in the encoded picture; and after decoding the plurality of initial

blocks, decoding a first block in the remaining of the plurality of blocks based on the decoded initial blocks.

4. The method of example 1, comprising: deciding, based on the determining, that a coded representation of the block being decoded excludes an indicator of use of a second coding technique; and parsing the coded representation based on the deciding. For example, in this parsing process, the decoder is successfully able to parse coded bitstream with the knowledge that fields related to the excluded coding technique will not be present in the bitstream.

5. A method of video processing, comprising: determining that an indicator associated with an encoded picture comprises an alternate type of reference picture signaling an encoding technique applied to the encoded picture is an IBC technique, wherein the alternate type of reference picture indicator is different from a long-term reference picture indicator and a short-term reference picture indicator, wherein a picture comprising the long-term reference picture indicator is stored in a memory for a first duration, and wherein a picture comprising the short-term reference picture indicator is stored in the memory for a second duration that is shorter than the first duration.

6. The method of example 5, comprising: obtaining a first indicator comprising the long-term reference picture indicator or the long-term reference picture indicator associated with a first picture; obtaining a second indicator comprising the alternate type of reference picture indicator associated with a second picture; avoiding decoding the second picture based on the first picture; avoiding decoding the first picture based on the second picture.

7. The method of example 6, comprising: obtaining a first indicator comprising the long-term reference picture indicator or the long-term reference picture indicator associated with a first picture; obtaining a second indicator comprising the alternate type of reference picture indicator associated with a second picture; skipping a scaling process of motion vectors associated with the first picture when decoding the second picture; and skipping the scaling process of motion vectors associated with the second picture when decoding the first picture.

8. The method of examples 1-7, comprising: when an indicator associated with a block of the encoded picture signals a use of the IBC technique, decoding the block without applying the affine decoding technique.

9. The method of example 8, comprising: obtaining a plurality of reference pictures used to decode a block of the encoded picture comprising a plurality of blocks; when at least one reference picture in the plurality of reference pictures comprises the encoded picture, decoding the block without applying the affine decoding technique.

10. The method of example 8, comprising: obtaining a plurality of reference pictures used to decode a block of the encoded picture comprising a plurality of blocks; in case that a block is associated with multiple sets of motion information, wherein the multiple sets of motion information comprise more than two sets of motion information, and each reference picture in the plurality of reference pictures is the encoded picture, decoding the block without applying the affine decoding technique.

11. The method of example 8, comprising: obtaining a plurality of reference pictures used to decode a block of the encoded picture comprising a plurality of blocks, the plurality of reference pictures comprising the encoded picture and another picture in the visual information; obtaining a first indicator signaling that the affine coding technique is

applied to the block; decoding the block using the affine decoding technique and the another picture in the visual information.

12. The method of example 11, wherein the encoded picture and the another picture in the visual information are identical, the method comprising: avoiding decoding the block using the affine decoding technique and the encoded picture. For example, avoiding may be performed by disabling the affine decoding technique and the encoded picture during the decoding process.

13. The method of examples 1-12, comprising: determining that an indicator associated with the encoded picture signals that the affine coding technique is applied to the encoded picture; and in case that the indicator associated with the encoded picture signals that the affine coding technique is applied to the encoded picture, decoding the encoded picture without applying the IBC decoding technique.

14. The method of example 13, comprising: obtaining one or more reference pictures used to decode a block of the encoded picture comprising a plurality of blocks and the indicator associated with the encoded picture; in case that the indicator associated with the encoded picture signals that the affine coding technique is applied to the encoded picture, assuming that the one or more reference pictures do not contain the encoded picture.

15. The method of example 13, comprising: obtaining one or more reference pictures used to decode a block of the encoded picture comprising a plurality of blocks and the indicator associated with the encoded picture; in case that the indicator associated with the encoded picture signals that the affine coding technique is applied to the encoded picture, assuming that at least one reference picture in the one or more reference pictures is different from the encoded picture.

16. The method of examples 1-7, comprising: obtaining a first indicator associated with the encoded picture and a second indicator associated with the encoded picture, the first indicator signaling that the affine coding technique is applied to the block, the second indicator signaling that the IBC technique is applied to the block; and decoding the block of the encoded picture comprising a plurality of blocks by applying the affine decoding technique and the IBC decoding technique to the block.

17. The method of example 16, comprising: obtaining an initial block in a plurality of blocks representing the encoded picture; decoding an initial motion vector of an initial control point associated with the initial block; obtaining a first block in the plurality of blocks representing the encoded picture, and a first indicator signaling that the IBC technique is applied to the first block; disabling motion vector scaling; and upon disabling motion vector scaling, creating a motion vector prediction (MVP) of a first control point associated with the first block based on the initial motion vector of the initial control point associated with initial block.

18. The method of examples 16-17, comprising: obtaining a first block in the plurality of blocks representing the encoded picture, and a first indicator signaling that the affine coding technique is applied to the first block; creating the MVP of the first control point associated with the first block based on a motion vector of a control point associated with the block belonging to a picture different from the encoded picture.

19. The method of examples 16-17, comprising: obtaining a plurality of blocks representing the encoded picture; obtaining a motion vector of a control point associated with a block in the plurality of blocks, wherein the motion vector

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is signaled in an integer precision; making a motion vector prediction based on the motion vector; and rounding the motion vector prediction to an integer-pixel precision.

20. The method of examples 16-17, comprising: obtaining a block in a plurality of blocks representing the encoded picture; deriving a motion vector at a control point associated with the block from a neighboring motion vector at a neighboring control point; and rounding the motion vector to an integer-pixel precision.

21. The method of examples 16-20, comprising: avoiding to conduct an Overlapped Block Motion Compensation (OBMC).

22. The method of examples 16-20, comprising: assuming that a predetermined component of a motion vector at a control point is a constant integer.

23. A method for encoding a visual information, comprising: encoding the visual information into a plurality of encoded pictures and a plurality of indicators signaling one or more encoding techniques applied, the plurality of indicators comprising an intra-block copy (IBC) technique indicator and an affine coding technique indicator, wherein a first block of a first picture associated with the visual information is encoded using the IBC technique and a second block of a second picture associated with the visual information is encoded using the affine coding technique, wherein the IBC technique uses a different block of the first picture to encode the first block of the first picture and the affine coding technique uses a third picture associated with the visual information to encode the second block.

24. The method of example 23, the IBC technique comprising: dividing the first picture into a plurality of blocks; encoding an initial block in the plurality of blocks; and upon encoding the initial block, encoding a first block in the plurality of blocks based on the initial block.

25. The method comprising: encoding an indicator in a plurality of indicators associated with an encoded picture in a plurality of encoded pictures, the indicator comprising an alternate type of reference picture indicator signaling an encoding technique applied to the encoded picture is an IBC technique, wherein the alternate type of reference picture indicator is different from a long-term reference picture indicator and a short-term reference picture indicator, wherein a picture comprising the long-term reference picture indicator is stored in a memory for a first duration, and wherein a picture comprising the short-term reference picture indicator is stored in the memory for a second duration that is shorter than the first duration.

26. The method of example 25, comprising: obtaining a first indicator comprising the long-term reference picture indicator or the long-term reference picture indicator associated with a first picture; obtaining a second indicator comprising the alternate type of reference picture indicator associated with a second picture; avoiding encoding the second picture based on the first picture; and avoiding encoding the first picture based on the second picture.

27. The method of example 25, comprising: encoding a first indicator comprising the long-term reference picture indicator or the long-term reference picture indicator associated with a first picture; encoding a second indicator comprising the alternate type of reference picture indicator associated with a second picture; skipping a scaling process of motion vectors associated with the first picture when encoding the second picture; and skipping the scaling process of motion vectors associated with the second picture when encoding the first picture.

28. The method of examples 23-27, comprising: applying IBC technique to a block of the first picture in the visual

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information to obtain an encoded block; encoding an indicator associated with the encoded picture as the IBC technique and skipping encoding the encoded block using the affine coding technique and encoding an indicator signaling the affine coding technique.

29. The method of example 28, comprising: encoding a plurality of reference pictures used to encode the block of the first picture; in case that at least one reference picture in the plurality of reference pictures comprises the first picture, skipping encoding an indicator of use of the affine coding technique.

30. The method of example 28, comprising: encoding a plurality of reference pictures used to encode a block of the first picture; when a block is associated with multiple sets of motion information, wherein the multiple sets of motion information comprise more than two sets of motion information, and each reference picture in the plurality of reference pictures is the first picture, skipping encoding an indicator signaling the affine coding technique.

31. The method of example 28, comprising: encoding a plurality of reference pictures used to encode a block of the first picture, the plurality of reference pictures comprising the first picture and another picture in the visual information; encoding a first indicator signaling that the affine coding technique is applied to the block; and applying the affine coding technique to the block of the first picture and the another picture in the visual information.

32. The method of example 31, wherein the encoded picture and the another picture in the visual information are identical, the method comprising: skipping applying the affine coding technique to the block of the first picture and the another picture in the visual information.

33. The method of examples 23-32, comprising: applying the affine coding technique to the second picture associated with the visual information; and skipping encoding an indicator comprising the IBC technique.

34. The method of example 33, comprising: encoding one or more reference pictures used to encode a block of the second picture, wherein the one or more reference pictures excludes the second picture; and encoding an indicator in the plurality of indicators to signal that the affine coding technique is used to encode the second picture.

35. The method of example 33, comprising: encoding one or more reference pictures used to encode a block of the second picture, wherein at least one reference picture in the one or more reference pictures is different from the second picture; and encoding an indicator in the plurality of indicators to signal that the affine coding technique is used to encode the second picture.

36. The method of examples 23-27, comprising: encoding a block of the first picture using the IBC technique and the affine coding technique; encoding a first indicator and a second indicator associated with the block of the first picture, the first indicator to signal that the IBC technique is used to encode the block and the second indicator to signal that the affine coding technique is used to encode the block.

37. The method of example 36, comprising: dividing the first picture into a plurality of blocks; encoding a motion vector of a control point associated with an initial block in the plurality of blocks; and encoding a motion vector of a control point associated with a first block in the plurality of blocks based on the motion vector of the control point associated with the initial block; and disabling motion vector scaling.

38. The method of examples 36-37, comprising: dividing the second picture into a second plurality of blocks and a third picture in the visual information into a third plurality

of blocks, wherein the third picture is different from the second picture; encoding a motion vector of a control point associated with a second block in second the plurality of blocks; and encoding a motion vector of a control point associated with a third block in the third plurality of blocks based on the motion vector of the control point associated with the second block.

39. The method of examples 36-37, comprising: dividing the first picture into a plurality of blocks; encoding a block in the plurality of blocks using the affine coding technique; encoding a motion vector of a control point associated with the block in the plurality of blocks using an integer precision; encoding a motion vector prediction based on the motion vector by rounding the motion vector prediction to an integer-pixel precision.

40. The method of examples 36-39, comprising: dividing the first picture into a plurality of blocks; encoding a block in the plurality of blocks using the affine coding technique; encoding a motion vector of a control point associated with the block in the plurality of blocks using an integer precision; encoding a motion vector of a point inside the block based on the motion vector of the control point associated with the block by rounding the motion vector of the point inside the block to the integer-pixel precision.

41. The method of examples 36-40, comprising: avoiding applying an Overlapped Block Motion Compensation (OBMC).

42. The method of examples 36-40, comprising: skipping encoding of a predetermined component of a motion vector at a control point.

43. The method of examples 1-42, comprising: transmitting an indicator between an encoder and a decoder, the indicator signaling whether and how to apply the affine coding technique for IBC technique coded blocks at a sequence level, a picture level, a slice level, a Coding Tree Unit (CTU), a region level, a CU level, or a PU level.

44. The method of example 43, comprising: transmitting the indicator at a Sequence Parameter Set (SPS), Picture Parameter Set (PPS), a Slice Header (SH), a tile group header, a CTU, a region, a CU or a PU.

45. A video processing apparatus comprising a processor configured to implement a method recited in any one or more of examples 1 to 44. In some embodiments, the video processing apparatus may be a video encoder. In some embodiments, the video processing apparatus may be a video decoder. In some embodiments, the video processing apparatus may be a video transcoder.

46. A computer readable medium having processor-executable code stored thereon, the code, upon execution, causing a processor to implement a method recited in any one or more of examples 1 to 44.

5. Example Implementations of the Disclosed Embodiments

FIG. 20 is a block diagram illustrating an example of the architecture for a computer system or other control device **2000** that can be utilized to implement various portions of the presently disclosed embodiments, including (but not limited to) methods **1600**, **1700**, **1800**, **1900**, **2200** and **2300**. In FIG. 20, the computer system **2000** includes one or more processors **2005** and memory **2010** connected via an interconnect **2025**. The interconnect **2025** may represent any one or more separate physical buses, point to point connections, or both, connected by appropriate bridges, adapters, or controllers. The interconnect **2025**, therefore, may include, for example, a system bus, a Peripheral Component Inter-

connect (PCI) bus, a HyperTransport or industry standard architecture (ISA) bus, a small computer system interface (SCSI) bus, a universal serial bus (USB), IIC (I2C) bus, or an Institute of Electrical and Electronics Engineers (IEEE) standard 674 bus, sometimes referred to as "Firewire."

The processor(s) **2005** may include central processing units (CPUs) to control the overall operation of, for example, the host computer. In certain embodiments, the processor(s) **2005** accomplish this by executing software or firmware stored in memory **2010**. The processor(s) **2005** may be, or may include, one or more programmable general-purpose or special-purpose microprocessors, digital signal processors (DSPs), programmable controllers, application-specific integrated circuits (ASICs), programmable logic devices (PLDs), or the like, or a combination of such devices.

The memory **2010** can be or include the main memory of the computer system. The memory **2010** represents any suitable form of random-access memory (RAM), read-only memory (ROM), flash memory, or the like, or a combination of such devices. In use, the memory **2010** may contain, among other things, a set of machine instructions which, when executed by processor **2005**, causes the processor **2005** to perform operations to implement embodiments of the presently disclosed embodiments.

Also connected to the processor(s) **2005** through the interconnect **2025** is a (optional) network adapter **2015**. The network adapter **2015** provides the computer system **2000** with the ability to communicate with remote devices, such as the storage clients, and/or other storage servers, and may be, for example, an Ethernet adapter or Fiber Channel adapter.

FIG. 21 shows a block diagram of an example embodiment of a mobile device **2100** that can be utilized to implement various portions of the presently disclosed embodiments, including (but not limited to) methods **1600**, **1700**, **1800** and **1900**. The mobile device **2100** can be a laptop, a smartphone, a tablet, a camcorder, or other types of devices that are capable of processing videos. The mobile device **2100** includes a processor or controller **2101** to process data, and memory **2102** in communication with the processor **2101** to store and/or buffer data. For example, the processor **2101** can include a central processing unit (CPU) or a microcontroller unit (MCU). In some implementations, the processor **2101** can include a field-programmable gate array (FPGA). In some implementations, the mobile device **2100** includes or is in communication with a graphics processing unit (GPU), video processing unit (VPU) and/or wireless communications unit for various visual and/or communications data processing functions of the smartphone device. For example, the memory **2102** can include and store processor-executable code, which when executed by the processor **2101**, configures the mobile device **2100** to perform various operations, e.g., such as receiving information, commands, and/or data, processing information and data, and transmitting or providing processed information/data to another device, such as an actuator or external display.

To support various functions of the mobile device **2100**, the memory **2102** can store information and data, such as instructions, software, values, images, and other data processed or referenced by the processor **2101**. For example, various types of RAM devices, ROM devices, Flash Memory devices, and other suitable storage media can be used to implement storage functions of the memory **2102**. In some implementations, the mobile device **2100** includes an input/output (I/O) unit **2103** to interface the processor **2101**

and/or memory **2102** to other modules, units or devices. For example, the I/O unit **2103** can interface the processor **2101** and memory **2102** with to utilize various types of wireless interfaces compatible with typical data communication standards, e.g., such as between the one or more computers in the cloud and the user device. In some implementations, the mobile device **2100** can interface with other devices using a wired connection via the I/O unit **2103**. The mobile device **2100** can also interface with other external interfaces, such as data storage, and/or visual or audio display devices **2104**, to retrieve and transfer data and information that can be processed by the processor, stored in the memory, or exhibited on an output unit of a display device **2104** or an external device. For example, the display device **2104** can display a video frame that includes a block (a CU, PU or TU) that applies the intra-block copy based on whether the block is encoded using a motion compensation algorithm, and in accordance with the disclosed embodiments.

In some embodiments, a video decoder apparatus may implement a method of video decoding in which the intra-block copy as described herein is used for video decoding. The method may be similar to the above-described methods **1600**, **1700**, **1800**, **1900**, **2200** and **2300**.

In some embodiments, a decoder-side method of video decoding may use the intra-block copy for improving video quality by determining whether a current block of the current picture is to be decoded using a motion compensation algorithm, and decoding, based on the determining, the current block by selectively applying an intra-block copy to the current block.

In other embodiments, a decoder-side method of video decoding may use the intra-block copy for improving video quality by determining whether a current block of the current picture is to be decoded using an intra-block copy, and decoding, based on the determining, the current block by selectively applying a motion compensation algorithm to the current block.

In some embodiments, the video decoding methods may be implemented using a decoding apparatus that is implemented on a hardware platform as described with respect to FIG. **20** and FIG. **21**.

Below are improvements measured by incorporating IBC into VVC test model (VTM)-1.0, which is a reference software for the video coding standard named Versatile Video Coding (VVC).

	Over VTM-1.0				
	Y	U	V	EncT	DecT
Class A1	-0.33%	-0.50%	-0.49%	162%	100%
Class A2	-0.96%	-1.17%	-0.77%	159%	98%
Class B	-0.94%	-1.14%	-1.34%	162%	102%
Class C	-1.03%	-1.58%	-1.92%	160%	101%
Class E	-1.48%	-1.46%	-1.80%	160%	104%
Overall	-0.95%	-1.19%	-1.31%	161%	101%
Class D	-0.57%	-0.73%	-0.91%	161%	100%
Class F (optional)	-20.25%	-20.15%	-20.93%	194%	95%
Class SCC 1080p	-52.94%	-53.26%	-53.37%	217%	74%

In the above table, “Y”, “U”, “V” represent colors in the YUV color encoding system which encodes a color image or video taking human perception into account. The EncT and DecT represent a ratio of the encoding and decoding time using the IBC compared to the encoding and decoding time without the IBC, respectively. Specifically,

$$\text{EncT} = \text{TestEncodingTime} / \text{anchorEncodingTime}$$

$$\text{DecT} = \text{TestDecodingTime} / \text{anchorDecodingTime}$$

The various classes, such as Class A1, Class A2, etc., represent a grouping of standard video sequences used in testing performance of various video coding techniques. The negative percentages under the “Y”, “U”, “V” columns represent bit-rate savings when IBC is added to VTM-1.0. The percentages under the EncT and DecT columns that are over 100% show how much the encoding/decoding with IBC is slower than encoding/decoding without IBC. For example, a percentage of 150% means that the encoding/decoding with IBC is 50% slower than the encoding/decoding without the IBC. The percentage below 100% shows how much the encoding/decoding with IBC is faster than encoding/decoding without the IBC. Two classes, class F and class SCC, highlighted in green in the table above, show that bit-rate savings exceed 3%.

From the foregoing, it will be appreciated that specific embodiments of the presently disclosed embodiments have been described herein for purposes of illustration, but that various modifications may be made without deviating from the scope of the present disclosure. Accordingly, the presently disclosed embodiments are not limited except as by the appended claims.

Implementations of the subject matter and the functional operations described in the present disclosure can be implemented in various systems, digital electronic circuitry, or in computer software, firmware, or hardware, including the structures disclosed in this specification and their structural equivalents, or in combinations of one or more of them. Implementations of the subject matter described in this specification can be implemented as one or more computer program products, i.e., one or more modules of computer program instructions encoded on a tangible and non-transitory computer readable medium for execution by, or to control the operation of, data processing apparatus. The computer readable medium can be a machine-readable storage device, a machine-readable storage substrate, a memory device, a composition of matter effecting a machine-readable propagated signal, or a combination of one or more of them. The term “data processing unit” or “data processing apparatus” encompasses all apparatus, devices, and machines for processing data, including by way of example a programmable processor, a computer, or multiple processors or computers. The apparatus can include, in addition to hardware, code that creates an execution environment for the computer program in question, e.g., code that constitutes processor firmware, a protocol stack, a database management system, an operating system, or a combination of one or more of them.

A computer program (also known as a program, software, software application, script, or code) can be written in any form of programming language, including compiled or interpreted languages, and it can be deployed in any form, including as a stand-alone program or as a module, component, subroutine, or other unit suitable for use in a computing environment. A computer program does not necessarily correspond to a file in a file system. A program can be stored in a portion of a file that holds other programs or data (e.g., one or more scripts stored in a markup language document), in a single file dedicated to the program in question, or in multiple coordinated files (e.g., files that store one or more modules, sub programs, or portions of code). A computer program can be deployed to be executed on one computer or on multiple computers that are located at one site or distributed across multiple sites and interconnected by a communication network.

The processes and logic flows described in this specification can be performed by one or more programmable

processors executing one or more computer programs to perform functions by operating on input data and generating output. The processes and logic flows can also be performed by, and apparatus can also be implemented as, special purpose logic circuitry, e.g., a FPGA or an ASIC.

Processors suitable for the execution of a computer program include, by way of example, both general and special purpose microprocessors, and any one or more processors of any kind of digital computer. Generally, a processor will receive instructions and data from a read only memory or a random access memory or both. The essential elements of a computer are a processor for performing instructions and one or more memory devices for storing instructions and data. Generally, a computer will also include, or be operatively coupled to receive data from or transfer data to, or both, one or more mass storage devices for storing data, e.g., magnetic, magneto optical disks, or optical disks. However, a computer need not have such devices. Computer readable media suitable for storing computer program instructions and data include all forms of nonvolatile memory, media and memory devices, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electronically erasable programmable read-only memory (EEPROM), and flash memory devices; magnetic disks, e.g., internal hard disks or removable disks; magneto optical disks; and compact disc, read-only memory (CD ROM) and digital versatile disc, read-only memory (DVD-ROM) disks. The processor and the memory can be supplemented by, or incorporated in, special purpose logic circuitry.

It is intended that the specification, together with the drawings, be considered exemplary only, where exemplary means an example. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Additionally, the use of “or” is intended to include “and/or”, unless the context clearly indicates otherwise.

While the present disclosure contains many specifics, these should not be construed as limitations on the scope of any embodiments or of what may be claimed, but rather as descriptions of features that may be specific to particular embodiments of the present disclosure. Certain features that are described in the present disclosure in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a subcombination or variation of a subcombination.

Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results. Moreover, the separation of various system components in the embodiments described in the present disclosure should not be understood as requiring such separation in all embodiments.

Only a few implementations and examples are described and other implementations, enhancements and variations can be made based on what is described and illustrated in the present disclosure.

What is claimed is:

1. A method of processing video data, comprising: determining, for a conversion between a current video block of a video and a bitstream of the video, that a first technique is applied for the current video block; and performing the conversion by applying the first technique and by excluding applying a second technique, wherein one of the first and the second techniques corresponds to an intra-block copy (IBC) technique, and the other of the first and the second techniques corresponds to a Decoder-side Motion Vector Refinement (DMVR) technique, wherein in response to the DMVR technique being applied during the conversion, a refined motion vector for the current video block is derived based on a cost calculation between prediction samples of the current video block in different reference pictures of the current video block, and wherein the method further comprises, in response to at least one of reference pictures of the current video block being a current picture which includes the current video block, excluding applying the DMVR technique for the current video block during the conversion.
2. The method of claim 1, wherein in response to the IBC technique being applied during the conversion, a prediction of the current video block is derived based on sample values from a same video slice with the current video block, and wherein the sample values are determined by block vectors.
3. The method of claim 1, wherein the cost calculation is based on a sum of absolute difference (SAD).
4. The method of claim 1, wherein the first technique corresponds to the IBC technique, and the second technique corresponds to the DMVR technique.
5. The method of claim 1, wherein the first technique corresponds to the DMVR technique, and the second technique corresponds to the IBC technique.
6. The method of claim 1, wherein the conversion includes encoding the current video block into the bitstream.
7. The method of claim 1, wherein the conversion includes decoding the current video block from the bitstream.
8. An apparatus for processing video data comprising a processor and a non-transitory memory with instructions thereon, wherein the instructions upon execution by the processor, cause the processor to: determine, for a conversion between a current video block of a video and a bitstream of the video, that a first technique is applied for the current video block; and perform the conversion by applying the first technique and by excluding applying a second technique, wherein one of the first and the second techniques corresponds to an intra-block copy (IBC) technique, and the other of the first and the second techniques corresponds to a Decoder-side Motion Vector Refinement (DMVR) technique, wherein in response to the DMVR technique being applied during the conversion, a refined motion vector for the current video block is derived based on a cost calculation between prediction samples of the current video block in different reference pictures of the current video block, and wherein the instructions upon execution by the processor, further cause the processor to, in response to at least one of reference pictures of the current video block being a current picture which includes the current video block, exclude applying the DMVR technique for the current video block during the conversion.

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9. The apparatus of claim 8, wherein in response to the IBC technique being applied during the conversion, a prediction of the current video block is derived based on sample values from a same video slice with the current video block, and wherein the sample values are determined by block vectors. 5

10. The apparatus of claim 8, wherein the cost calculation is based on a sum of absolute difference (SAD).

11. The apparatus of claim 8, wherein the first technique corresponds to the IBC technique, and the second technique corresponds to the DMVR technique. 10

12. The apparatus of claim 8, wherein the first technique corresponds to the DMVR technique, and the second technique corresponds to the IBC technique. 15

13. A non-transitory computer-readable storage medium storing instructions that cause a processor to:

determine, for a conversion between a current video block of a video and a bitstream of the video, that a first technique is applied for the current video block; and perform the conversion by applying the first technique and by excluding applying a second technique, wherein one of the first and the second techniques corresponds to an intra-block copy (IBC) technique, and the other of the first and the second techniques corresponds to a Decoder-side Motion Vector Refinement (DMVR) technique, 20

wherein in response to the DMVR technique being applied during the conversion, a refined motion vector for the current video block is derived based on a cost calculation between prediction samples of the current video block in different reference pictures of the current video block, and 25

wherein the instructions further cause the processor to, in response to at least one of reference pictures of the current video block being a current picture which includes the current video block, exclude applying the DMVR technique for the current video block during the conversion. 30

14. The non-transitory computer-readable storage medium of claim 13, 35

wherein in response to the IBC technique being applied during the conversion, a prediction of the current video block is derived based on sample values from a same video slice with the current video block, and wherein the sample values are determined by block vectors, 40

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wherein the cost calculation is based on a sum of absolute difference (SAD), and

wherein the first technique corresponds to the IBC technique, and the second technique corresponds to the DMVR technique, or wherein the first technique corresponds to the DMVR technique, and the second technique corresponds to the IBC technique.

15. A method of storing a bitstream of a video, comprising: 45

determining, for a current video block of the video, that a first technique is applied for the current video block; generating the bitstream from the current video block by using the first technique and by excluding applying a second technique, and 50

storing the bitstream in a non-transitory computer-readable recording medium, 15

wherein one of the first and the second techniques corresponds to an intra-block copy (IBC) technique, and the other of the first and the second techniques corresponds to a Decoder-side Motion Vector Refinement (DMVR) technique, 20

wherein in response to the DMVR technique being applied during the conversion, a refined motion vector for the current video block is derived based on a cost calculation between prediction samples of the current video block in different reference pictures of the current video block, and 25

wherein the method further comprises, in response to at least one of reference pictures of the current video block being a current picture which includes the current video block, excluding applying the DMVR technique for the current video block during the conversion. 30

16. The method of claim 15, wherein in response to the IBC technique being applied during the generating, a prediction of the current video block is derived based on sample values from a same video slice with the current video block, and wherein the sample values are determined by block vectors, 35

wherein the cost calculation is based on a sum of absolute difference (SAD), and

wherein the first technique corresponds to the IBC technique, and the second technique corresponds to the DMVR technique, or wherein the first technique corresponds to the DMVR technique, and the second technique corresponds to the IBC technique. 40

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